# Education Subsidy as Insurance for Fertility Choices*

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February 2025

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**Abstract**

This paper analytically and quantitatively shows that the income-tested college subsidies, which prevail across countries, provide insurance for fertility choices. I develop this argument by constructing a lifecycle Aiyagari model with fertility and college enrollment choices, in which altruistic parents make asset transfers to their children to support their college education. I calibrate the model by using Japanese household panel data and then validate the model by showing that it replicates several measures of fertility elasticities consistent with empirical estimates. For parents, having more children increases the likelihood that they may be unable to afford their children's education if they face bad income shocks in the future—or that they end up with significantly low lifetime consumption to finance their education while facing bad income shocks—deterring them from having more children under income uncertainty. Counterfactual experiments using the model demonstrate that the income-tested college subsidy provides insurance against this risk associated with having a (or another) child. Notably, the resulting fertility responses amplify the macroeconomic effects of the college subsidy—including reducing income inequality, improving welfare, and increasing education and income levels—through intergenerational linkages and GE feedback. The transitional dynamics highlight this amplification along the transition path.

Keywords: Fertility, incomplete market, overlapping generations, college subsidy
JEL codes: C68, I28, J13, J24

* I am grateful to Minsu Chang, James Graham, Jeremy Greenwood, Yusuke Ishihata, Akihisa Kato, Sagiri Kitao, Tomoaki Kotera, Dirk Krueger, Minjoon Lee, Kazushige Matsuda, Toshihiko Mukoyama, Makoto Nakajima, Tomoyuki Nakajima, Makoto Nirei, Yena Park, Ponpoje Porapakkarm, Xavier Ragot, Víctor Ríos-Rull, Cezar Santos, Kamila Sommer, Yuta Takahashi, Satoshi Tanaka, Kazuhiro Teramoto, Hanna Wang, Tomoaki Yamada, Shintaro Yamaguchi, Minchul Yum, and participants at numerous seminars and conferences for their many helpful comments. This study is financially supported by the Japan Society for the Promotion of Science (23KJ0374).

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# 1 Introduction

Government subsidies for college education are prevalent across countries. While they are often regarded as effective tools for reducing income inequality and fostering economic growth, it is crucial to understand the specific margins through which these policies achieve their goals, as well as their potential side effects, which are essential for designing effective policies. In this spirit, previous studies highlight key margins affecting the costs and benefits of college subsidies. These include the general equilibrium (GE) feedback of the policy through an increased supply of college-educated workers (e.g., Krueger and Ludwig, 2016), the crowding-out of parental transfers and student labor supply that may dampen policy effects (Abbott, Gallipoli, Meghir, and Violante, 2019), and the complementarity between college subsidies and policies promoting pre-college human capital accumulation that affects college completion success (Krueger, Ludwig, and Popova, 2024).

The central contribution of this paper is to find a new key margin and channel in assessing the macroeconomic implications of college subsidies: fertility choices. This contribution rests on three key components. First, I analytically and quantitatively show that income-tested college subsidies, which prevail across countries, serve as partial insurance for fertility choices in the following sense. For parents, having more children increases the likelihood that they may be unable to afford their children's education if they face bad income shocks in the future—or that they end up with significantly low lifetime consumption to finance their education while facing bad income shocks—deterring them from having more children under income uncertainty: the income-tested college subsidy provides insurance against this risk associated with having a (or another) child. Second, I find that the resulting fertility responses amplify the macroeconomic effects of the college subsidy—such as inequality reduction, welfare improvement, and increasing income levels—through the associated general equilibrium (GE) effects and intergenerational linkages. These findings build on the third key component: this study is the first to construct a GE lifecycle model that integrates fertility and college enrollment choices.

I start with a simple analytical model of fertility choices with uninsurable income risk to highlight an insurance effect of income-tested subsidies on fertility. In this model, ex-ante identical households make fertility choices under income uncertainty, with spending on children assumed to be fixed and exogenously given, and income levels are realized after fertility choices are made. Having children can ex-post become very costly in utility terms following a bad income realization because, if they have more children, a bad income shock results in significantly lower consumption (or, lower spending on children if it were endogenous) compared to having fewer children, leading to higher marginal utilities. However, parents cannot "reduce" the number of children after facing a bad income shock because fertility choices are irreversible. Therefore, income uncertainty may deter parents from having another child, lowering fertility rates—an empirically supported relationship in previous studies.[1] I consider an income-tested subsidy that redistributes income from households with positive (good) income shocks to those with negative (bad) shocks, ensuring that the expected value of the subsidy received is zero. This design rules out the subsidy's effects on fertility driven by lower expected monetary costs of children

[1] See, for example, Modena et al. (2014); Guner et al. (2024). Although parents may choose to postpone childbearing to a later stage in life, this ultimately results in lower (completed) fertility rates, as the ability to bear children declines with age (Sommer, 2016).

(or the lower "price" of children), which I refer to as the *price effect*. I show that despite the absence of this effect, income-tested subsidies still increase fertility: they smooth marginal utilities conditional on having children, thereby lowering the expected marginal costs of children, even though the expected monetary costs remain unchanged. I refer to this fertility increase as the *insurance effect* of the subsidy.

To evaluate the quantitative importance of this insurance effect relative to other potential effects, which is informative for policy design, and to examine the macroeconomic effects associated with these fertility margins, I develop a lifecycle Aiyagari model (i.e., an incomplete market, GE lifecycle model) with several key elements: (1) fertility choices about how many children to have, (2) college enrollment choices, (3) intergenerational linkages between parents and children through asset transfers and exogenous transmission of innate abilities and tastes for education, and (4) imperfect substitutability among workers with different educational backgrounds in the aggregate production function.

The model is calibrated using the Japanese Panel Survey of Consumers (JPSC). Japan provides an interesting case study because (1) the Japanese government has provided less financial support for higher education than other countries, (2) descriptive evidence suggests that the cost of college education—primarily borne by parents rather than students or the government—poses a significant barrier to fertility, and (3) the government recently introduced an income-tested college subsidy as a pro-natal policy.[2]

The calibrated model replicates key empirical moments, including those related to the income process, the distribution of the number of children, the amount of asset transfers from parents to college students, and the intergenerational persistence of educational status. Importantly, the model also replicates several measures of fertility elasticities as non-targeted moments: (1) fertility elasticity with respect to government cash transfers to households with children, as reported in previous empirical studies (e.g., Milligan, 2005; Malkova, 2018), (2) cross-country estimates of fertility elasticity with respect to college subsidies, and (3) fertility differentials between college- and high school–graduate parents.

In the benchmark model, college students can access government-provided student loans that mimic the actual loans available in Japan, which are income- and ability-tested, while grants are unavailable, reflecting the situation in Japan before 2020.[3] I then conduct several experiments using the model, including the introduction of income-tested grants implemented in 2020. These grants provide cash transfers to college students from households in the bottom 15% of the income distribution, and per-student grants amount to two-thirds of the average expenses of college students.

The main findings are summarized as follows. First, the direct effects of introducing the income-tested subsidy—defined as the effects of the subsidy while holding prices, tax rates, and household distribution constant—indicate a 2.1% increase in fertility. A decomposition analysis reveals that insurance effects play a more significant role than price effects: 62% of the direct effects on fertility are attributed to the insurance effect, while 38% are due to the price effect.

Next, I also find that these fertility responses amplify the policy's effects on inequality

[2] Appendix C. provides further details on these backgrounds in Japan.

[3] This model does not allow students to access financial aid (grants or loans) provided by institutions besides the government (e.g., colleges or other private entities). In 2018, among students in four-year universities who used any financial aid, 83.8% used only government-provided aid, 8.6% used only non-government-provided aid, and 7.6% used both, according to the Student Life Survey (SLS, 2018).

reduction, welfare improvement, and raising education and output levels in the long run through the following mechanism. First, the positive effect on fertility is more significant among college graduates since their children are more likely to attend college and are more likely to be eligible for the subsidy. Given intergenerational linkages via abilities, tastes, and assets, a higher fertility among college graduates implies that the children born due to the policy are more likely to attend college. Therefore, the policy further increases college enrollment rates through this fertility channel in the long run. A higher college enrollment rate reduces the skill premium and contributes to greater accumulation of aggregate human capital, thereby expanding the tax base and mitigating the increase in tax rates needed to finance the policy. These effects (i.e., a higher probability of attending college, higher expected income, lower inequality, and lower taxes), induced by the fertility responses and associated GE effects, generate more significant welfare gains for newborn agents under the veil of ignorance. This amplification is highlighted by comparing the results with those under an *exogenous fertility* setup, both at the steady state and during transitional dynamics.

The marginal expansion of the policy by setting a higher income threshold further increases fertility among college graduates, and this fertility response contributes to a higher college enrollment rate via intergenerational linkages. However, this expansion reduces the fertility of high school graduates because it significantly increases the intergenerational mobility of education and results in a "quantity-quality trade-off" of children. Also, the positive effects on output in expansion diminish since the broader coverage significantly crowds out labor supply and savings among households with children.

To summarize, I show that (1) income-tested subsidies provide insurance for fertility choices under income uncertainty and (2) the resulting fertility responses amplify the macroeconomic effects of the policy. Although the model is applied to Japanese data, the underlying mechanisms, highlighted through both analytical and quantitative models, are not unique to Japan but offer insights applicable to other countries as well.

## 2 Related Literature

This paper contributes to several strands of literature. The first is the macroeconomic analysis of college education subsidies (e.g., Krueger and Ludwig, 2016; Abbott, Gallipoli, Meghir, and Violante, 2019; Matsuda and Mazur, 2022; Krueger, Ludwig, and Popova, 2024). This study is the first to incorporate fertility choices into an otherwise standard framework in the literature (i.e., an incomplete market, GE lifecycle model with college enrollment choices and intergenerational linkages). Based on this extended framework, this study contributes to the literature by finding the critical roles of fertility margins in understanding the macroeconomic roles of income-tested college subsidies.

Second, this paper builds on macroeconomic studies based on the quantity-quality trade-off framework.[4] In particular, this study is closely related to three papers in the literature that consider the macroeconomic effects of education policies using models with the quantity and quality of children: De la Croix and Doepke (2004), Zhou (2022), and

[4]For example, De La Croix and Doepke (2003); De la Croix and Doepke (2004); Manuelli and Seshadri (2009); Greenwood, Guner, Kocharkov, and Santos (2016); Cordoba, Liu, and Ripoll (2019); Daruich and Kozlowski (2020); Zhou (2022); Kim, Tertilt, and Yum (2023).

Kim, Tertilt, and Yum (2023). These studies highlight the important roles of (1) fertility differentials in understanding the macroeconomic effects of private and public schooling regimes (De la Croix and Doepke, 2004), (2) the effectiveness of public education expenditures in boosting fertility compared to other pro-natal policies, along with their broader macroeconomic implications (Zhou, 2022), and (3) the externalities of parental investments in evaluating pro-natal policies and education taxes, as well as in understanding the underlying causes of low fertility in Korea (Kim, Tertilt, and Yum, 2023). I contribute to the literature by studying and highlighting the insurance role of education subsidies in fertility choices and their macroeconomic implications. From a modeling perspective, the closest work is Daruich and Kozlowski (2020). They develop a partial equilibrium lifecycle model with college enrollment, fertility, and inter-vivos transfers to investigate the role of fertility choices and family transfers in explaining intergenerational mobility in the U.S. Building on their work, this study extends their framework by adopting a GE approach to examine the macroeconomic implications of education policies.

Third, this study is also related to the literature on fertility choices in incomplete market models and, in general, on the implications of heterogeneous family structures under incomplete markets.[5] Previous studies demonstrate that having a child can be considered making "consumption commitments," and that income volatility thus makes households hesitate to have children, leading to a lower fertility (Sommer, 2016; Guner, Kaya, and Sánchez-Marcos, 2024).[6] This study contributes to the literature by examining policies that can potentially insure against the risks associated with having a child and highlights the insurance effects of income-tested college subsidies on fertility choices.

Finally, this study contributes to the literature on pro-natal policies using structural models[7] by adding information about the effects of college subsidies, a novel pro-natal policy implemented in Japan. Compared with typical pro-natal transfers such as baby bonuses, the notable feature of education subsidies is that it increases the average human capital by promoting skill acquisition.[8] By examining this novel policy in Japan, this study provides insights into countries considering countermeasures against macroeconomic challenges of low fertility.

[5]For example, Cubeddu and Ríos-Rull (2003); Schoonbroodt and Tertilt (2014); Vandenbroucke (2014); Santos and Weiss (2016); Sommer (2016); Bacher, Grübener, and Nord (2024); Coskun and Dalgic (2024); Guner, Kaya, and Sánchez-Marcos (2024).

[6]This effect is particularly pronounced in the early stages of life, leading to delayed fertility. Since the ability to reproduce declines with age, delayed fertility ultimately results in lower fertility rates (Sommer, 2016). Similarly, Santos and Weiss (2016) show that an increase in income volatility leads to delays in marriage, as marriage often entails a consumption commitment in the form of raising children. Coskun and Dalgic (2024) develop an overlapping generations model incorporating the quantity-quality trade-off and highlight the importance of accounting for gender differences in income volatility, as well as in income levels, within households in understanding the cyclical behavior of fertility.

[7]Previous studies examine the effects of childcare subsidization (e.g., Bick, 2016), cash transfers (e.g., Kim, Tertilt, and Yum, 2023; Nakakuni, 2024), both of them (e.g., Hagiwara, 2021; Zhou, 2022), parental leave policies (e.g., Erosa, Fuster, and Restuccia, 2010; Yamaguchi, 2019; Wang, 2022; Kim and Yum, 2023), and tax reform (e.g., Jakobsen, Jørgensen, and Low, 2022).

[8]Previous studies show that pro-natal transfers would lower aggregate human capital because they make parents shift from the "quality" toward "quantity" of children (e.g., Zhou, 2022; Kim et al., 2023). Indeed, Appendix L. shows that the income-tested grants would lead to greater aggregate human capital and output than pro-natal transfers conducted in an expenditure-neutral way.

# 3 An Illustrative Model

This section constructs a simple analytical model of fertility choices with uninsurable income risk to highlight the insurance effect of income-tested subsidies on fertility.[9]

**Setup:** Consider a decision problem where ex-ante identical households choose the number of children $n \geq 0$ and consumption $c > 0$, facing uninsurable income risk, to maximize their expected utility, $\mathbb{E}U(n, c) = b(n) + \mathbb{E}V(c)$. For simplicity, this model assumes that the number of children is a continuous variable and needs to be positive. The household first chooses the number of children $n$, before an income level $I$ is realized. After making the fertility choices, an income level is realized, which is drawn from a distribution $F(I)$. The price of the consumption goods is normalized to one, and having children takes a fixed cost $\underline{a}$, net of the government subsidy $\bar{g}(I)$. Let $a(I) = \underline{a} - \bar{g}(I)$ be the (net) unit costs of having a child. The household problem is formulated as follows:

$$\begin{aligned}
&\max_{n>0} \{b(n) + \mathbb{E}_I V(c)\} \\
\text{s.t. } & c + n \cdot a(I) = I, \\
& a(I) = \underline{a} - \bar{g}(I), \\
& I \sim F(I).
\end{aligned}$$

**The first-order condition:** The first-order condition implies that the optimal choices $(n, c)$ satisfy the following equation:

$$b'(n) = \mathbb{E}[a(I) \cdot V'(c)], \tag{1}$$

where a prime denotes the first-order derivative of a function with respect to its argument. The left-hand side of (1) indicates the marginal utility of having children. The right-hand side indicates the expected marginal costs of having children, which consists of the unit costs of children ($a(I)$) and the marginal utility from consumption ($V'(c)$). When they make fertility choices, the marginal costs of having children are uncertain because the amount of subsidy received and consumption level are uncertain; either source of uncertainty is income uncertainty.

**Assumptions:** To understand the effects of the income-tested subsidy on fertility, I make the following assumptions on functional forms.

First, the utility function of children $b(n)$ takes the form of the constant relative risk aversion utility function, $n^{1-\sigma}/(1-\sigma)$, with some positive value $\sigma > 0$, and the utility function for consumption $V$ satisfies the following properties: (1) $V' > 0$, (2) $V'' < 0$, and (3) $V''' > 0$. Second, the subsidy function $\bar{g}(I)$ takes the form $g(I; \varepsilon) = (\mathbb{E}I - I) \cdot \varepsilon$, where $\varepsilon \geq 0$ is a policy parameter that governs the degree of the redistribution. Note that this subsidy function $g(I; \varepsilon)$ implies that (1) the subsidy is income-contingent (or income-tested), where households with below-average income realizations receive a

[9] Appendix B. presents a special case of this model and shows that an increase in income volatility, represented by a mean-preserving spread of the income distribution, reduces fertility—consistent with empirical and quantitative studies (e.g., Sommer, 2016; Guner, Kaya, and Sánchez-Marcos, 2024).

positive subsidy, while those with above-average income realizations receive a negative subsidy or incur a "tax," and (2) the expected cost of children remain constant at $\underline{a}$ because the expected value of the subsidy is zero for any $\varepsilon$ (i.e., $\mathbb{E}g(I;\varepsilon) = \mathbb{E}(\mathbb{E}I - I) \cdot \varepsilon = 0$).

The first-order condition implies that the optimal fertility choice is characterized as:

$$n = \{\mathbb{E}[a(I) \cdot V'(c)]\}^{-\frac{1}{\sigma}}$$

This means that the fertility rate decreases with the expected marginal costs of having children, which comprise both the monetary costs of children and the utility costs associated with foregone consumption. For convenience, from now on, I express the optimal choices for consumption and fertility as functions of the parameter $\varepsilon$, $c(\varepsilon)$ and $n(\varepsilon)$. I define the parameter space for $\varepsilon$ as $\mathcal{E} \equiv \{\varepsilon \mid 1 - \varepsilon \cdot n(\varepsilon) > 0 \ \& \ \varepsilon \geq 0\}$, which ensures that higher (lower) income realizations correspond to higher (lower) consumption (i.e., $\partial c/\partial I > 0$ and thus $\partial V'(c)/\partial I < 0$).

**Effects of income-tested subsidies on fertility:** To understand the effects of income-tested subsidies on fertility, I compare the optimal fertility choice based on an economy with some small $\varepsilon \in \mathcal{E}$ with that with $\varepsilon = 0$, $n(\varepsilon)$ and $n(0)$. The ratio between $n(\varepsilon)$ and $n(0)$ can be written as follows:

$$\frac{n(\varepsilon)}{n(0)} = \left\{\frac{\mathbb{E}V'(c(\varepsilon))}{\mathbb{E}V'(c(0))} - cov(g(I;\varepsilon), V'(c(\varepsilon))) \cdot B\right\}^{-\frac{1}{\sigma}} > 1, \tag{2}$$

where $B = (\underline{a} \cdot \mathbb{E}V'(c(0)))^{-1} > 0$, and the inequality (2) implies that the fertility rate with an income-tested subsidy with $\varepsilon \in \mathcal{E}$ is greater than that without the subsidy:

$$n(\varepsilon) > n(0)$$

The derivation of inequality (2) is provided in Appendix D.. Here, I explain the intuition behind why this income-tested subsidy increases fertility.

First, having an additional child is costly in utility terms because it requires households to forgo consumption that they could otherwise enjoy. The marginal costs of having children consist of the expected monetary costs and the expected marginal utility from consumption, which represent the opportunity costs of having children. Recall that the expected monetary costs are identical across different subsidy schemes characterized by different $\varepsilon \in \mathcal{E}$ since $\mathbb{E}g(I;\varepsilon) = \mathbb{E}(\mathbb{E}I - I) = 0$. Therefore, it is only the expected marginal utility from consumption ($\mathbb{E}V'(c(\varepsilon))$) that makes a difference across different subsidy schemes in this setup.

If they have more children, a bad income shock is more costly in marginal utility because, holding the monetary costs of children constant, it leads to lower consumption. Given that having children is an irreversible choice, households cannot adjust the number of children after observing an income level. If a subsidy $g(I;\varepsilon)$ is available, such costs of having children in marginal utility from consumption become lower because they can receive some cash transfers after facing a bad income shock. It is also true that the subsidy imposes some costs for those with higher income realizations. However, the benefit of this insurance exceeds these costs, give that the third-order derivative of the utility function of consumption is positive (i.e., $V''' > 0$).

To recap, the intuition behind the higher fertility due to the income-tested subsidies is that (1) having children can be significantly costly ex-post in terms of foregone consumption and the resulting higher marginal utility, (2) an income-tested subsidy smooths the marginal utility to some extent, which we can interpret it as partial insurance against the high (marginal) utility costs of children induced by income shocks, and (3) the benefit of insurance exceeds the costs of paying "tax" upon a good income realization if $V''' > 0$.

**Next steps:** This section presents a simple analytical model and demonstrates that income-tested subsidies can positively impact fertility by providing insurance for fertility choices under income uncertainty. This raises further questions to explore: (1) to what extent do real-world income-tested education subsidies provide such insurance effects? and (2) what are the implications of the resulting fertility responses for other macroeconomic ingredients, such as aggregate output and income distribution? The current model is too simple to answer these questions in that it abstracts several critical aspects, such as (i) more realistic income process over the lifecycle, (ii) saving decisions as a self-insurance against income risk, (iii) endogenous choices on how much money to spend on children, and (iv) market production and the general equilibrium setup, which allows us to examine the macroeconomic implications. The next section thus constructs a quantitative macroeconomic model that incorporates those ingredients.

# 4 Quantitative Model

I incorporate fertility choices into an otherwise standard framework in the macroeconomic literature of college subsidies: an incomplete market, GE lifecycle model with college enrollment choices and intergenerational linkages. Section 4.1 provides an overview of the lifecycle of this economy. Section 4.2 elaborates on the preliminaries of the model, and then Sections 4.3–4.5 formulate household's problems, the firm's problem, and the government budget constraint. Section 4.6 discusses the equilibrium of this economy.

## 4.1 Overview of the lifecycle

Fig 1 represents the households' lifecycle in this model. Time is discrete, and one period corresponds to two years in this model. Let $j$ denote the agents' age. They live with their parents until they graduate from high school at the beginning of age $j = J_E (:= 18)$; until then, they make no decisions. After graduating high school, they choose whether to attend college or enter the labor market after graduating high school, represented as a node "Grad.HS" in the figure. If they do not enroll in college, they enter the labor market as a high school graduate. If they choose to attend college, it takes four years (two model periods) to complete, and they enter the labor market as a college graduate after graduation, represented as a node "Grad.CL" in the figure.[10]

After completing their education, they enter the labor market. At the beginning of age $j = J_F (:= 30)$, they make fertility choices by choosing how many children they have,

[10] This model does not consider the possibility of dropping out from college because, as the dropout rate is insignificant in Japan. For example, the dropout rate was 2.5% in 2021, according to the Ministry of Education, Culture, Sports, Science and Technology.

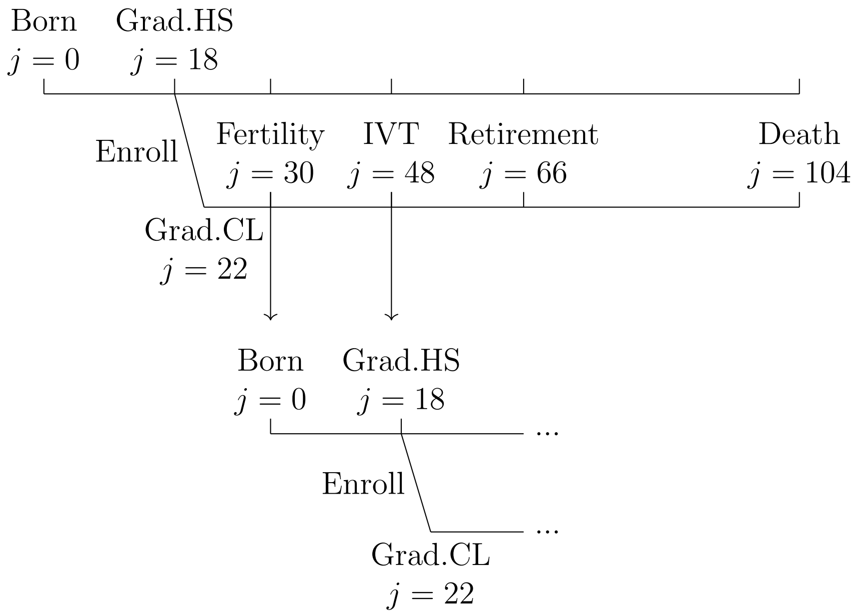


Fig 1: Model's lifecycle.

$n \in \{0, 1, 2, 3, 4\}$. The timing of the fertility choice is common for high school and college graduates. The lifecycle of a new cohort starts at this point (provided that the fertility rate is positive), represented in the bottom half of Fig 1. After their children (if they have any) graduate from high school, corresponding to the beginning of the age $j = J_{IVT} (:= 48)$ for parents, they decide how much money to transfer to their children. This IVT decision affects the children's college enrollment choices at the node "Grad.HS." Households retire from the labor market at the beginning of age $j = J_R (:= 66)$. After that period, they face mortality risks; every period, a certain fraction of them is hit by exogenous mortality shocks and exits from the economy. They can live for $J$ $(:= 104)$ years at the longest, and they exit the economy after age $J$.

Dividing the model's lifecycle into several stages, summarized in Table 1, helps us understand its structure; according to these stages, I formulate household problems in Section 4.3. Also, to provide a whole picture of the model's lifecycle, I provide lists of choices and state variables in Tables 2 and 3 in advance, including their relevant stages. Each variable is explained in detail in the following subsections.

| Stages | Corresponding age |
|---|---|
| (1) Education stage | $18(-21)$ |
| (2) Early working stage (before fertility stage) | $18(22) - 29$ |
| (3) Fertility stage | 30 |
| (4) Middle working stage (with children if applicable) | $30 - 47$ |
| (5) Inter vivos transfers (IVTs) stage | 48 |
| (6) Late working stage (after children become independent) | $48 - 65$ |
| (7) Retirement stage | $66-$ |

Table 1: The lifecycle stages and the corresponding age. *Note*: Numbers in parentheses indicate the corresponding age for those who choose to attend college.

| Notation | Age/Period | Description |
|---|---|---|
| Throughout the entire stages | | |
| $c > 0$ | 18− | Consumption |
| $l \in [0, 1]$ | 18− | Leisure |
| $a' \in [\underline{A}, \infty)$ | 18− | Asset |
| Education stage | | |
| $e \in \{0, 1\}$ | 18 | Education: college (=1) or high-school (=0) |
| Fertility, middle working, & IVT stages | | |
| $n \in \{0, 1, 2, 3, 4\}$ | 30 | The number of children (fertility) |
| $q > 0$ | $30 - 47$ | Expenditures on children |
| $a_{IVT} \geq 0$ | 48 | Inter vivos transfers |

Table 2: Summary of the choice variables over the life stages. *Note*: The borrowing limit, $\underline{A}$, varies over the lifecycle, as discussed in the following subsections.

| Notation | Age/Period | Description |
|---|---|---|
| Throughout the entire stages | | |
| $j$ | 18− | Age |
| $a$ | 18− | Asset |
| Until retirement | | |
| $z$ | $18(22) - 65$ | Labor productivity $\sim$ AR(1) |
| $e$ | $18(22) - 65$ | Education: college (=1) or high-school (=0) |
| $h$ | $18 - 65$ | Innate ability |
| Education stage | | |
| $\phi$ | 18 | Psychic costs of college enrollment |
| $I$ | $18(-21)$ | Household income |
| Middle working stage | | |
| $n$ | $30 - 48$ | The number of children |
| Inter vivos transfers stage | | |
| $h_k$ | 48 | Children's innate ability |
| $\phi_k$ | 48 | Children's psychic costs of college education |

Table 3: Summary of the state variables over the life stages. *Note*: Numbers in parentheses indicate the corresponding age for those who choose to attend college.

## 4.2 Preliminaries

**Demographics:** Every period, a mass of new cohorts enters the economy. The size of the new cohort is endogenously determined according to fertility choices of cohorts in the fertility stage. The age distribution of this economy is thus determined by (1) households' fertility choices, which are endogenous, and (2) mortality risks after retirement, which are exogenous. Let $\zeta_{j,j+1}$ denote the probability of surviving at age $j+1$ conditional on surviving at age $j$ for each $j \in \{J_R, ..., J\}$ with $\zeta_{J,J+1} = 0$.

**Intergenerational Linkages and Initial Endowments:** After graduating high school (at the beginning of age 18), agents are endowed with a triple $(a_{IVT}, h, \phi)$: (1) assets transferred from their parents ($a_{IVT}$), (2) innate ability ($h$), which governs education returns in future earnings, and (3) psychic costs of college education ($\phi$). They draw the innate ability from a distribution $g^h_{h_p}$, varying with the parents' innate ability level $h_p$. They also draw the psychic costs from a distribution $g^\phi_{h,e_p}$, varying with the student's innate ability $h$ and parent's education $e_p$.

**Preferences:** Every period, they draw utility from consumption $c$ and leisure $l$ according to a utility function $u(c, l)$. They discount future utility by a factor $\beta$. At age $j = J_F$, they choose the number of children they have, denoted by $n \in \{0, 1, 2, 3, 4\}$. They draw additional utility from having children in several ways. First, they draw utility from expenditures $q$ per child until the children graduate from high school, captured by a utility function $v(q)$. The utility $v(q)$ is discounted by a function $b(n)$, increasing and concave in the number of children $n$. Therefore, having more children (i.e., a higher $n$) also increases utility, holding other factors constant. Further, based on altruistic motives, parents draw utility by transferring assets to their children after the children graduate from high school. More specifically, parents consider children's expected lifetime utility in their education stage, with a discount rate $\lambda_a \cdot b(n)$, where $\lambda_a$ represents the altruistic discount factor.

**Costs of Children:** Having children is costly in terms of both money and time. First, raising $n$ children requires $n \cdot q$ units of money, where $q$ represents the per-child expenditure. Also, they will make an additional expenditure $n \cdot a_{IVT}$ upon high school graduation of their children.[11] In addition to the monetary costs, having a child requires $\kappa$ units of time until the child graduates from high school and becomes independent.

**Labor Earnings:** Households choose hours worked to earn income. The labor earnings of a household are given by $w_e \eta_{j,z,e,h}(T - l)$; it is a product of the market wage $w_e$, labor productivity $\eta_{j,z,e,h}$, and hours worked $(T - l)$. The market wage $w_e$ depends on the education level $e \in \{HS, CL\}$, where $e = HS$ and $e = CL$ represent high school and college graduates. Here, $T$ denotes the disposable time that can be devoted to work or leisure where $T = 1 - \kappa \cdot n$ if they have $n$ children who have not graduated high school and $T = 1$ otherwise. Labor productivity depends on age $j$, idiosyncratic component $z$, education level $e$, and innate ability $h$. The idiosyncratic component $z$ follows a Markov process $\pi_z(z, z^-)$, where $z^-$ denotes the component in the previous period.

[11] Children within a household are homogeneous, as assumed in the literature (e.g., Abbott, Gallipoli, Meghir, and Violante, 2019). Thus, education spending and IVT do not vary among children within the household.

**Financial Markets:** Financial markets are incomplete due to the lack of state-contingent claims. Households can self-insure against negative labor productivity shocks by savings, accruing interest payments at a rate of $r$. Households in the working stages can borrow at rate $r^- = r + \iota$ where $\iota > 0$ (i.e., borrowers incur the overseeing costs $\iota$) up to a borrowing limit $\underline{A}$. In contrast, retired households are not allowed to borrow, which is a common assumption in the literature. In addition, eligible students have access to student loans subsidized by the government, which entails the interest rate of $r^s = r + \iota_s$. This loan is income- and ability-tested, and eligible students can borrow up to a limit $\underline{A}_s$.

## 4.3 Household Problems

This section discusses and formulates household problems at each stage, as summarized in Table 1. The first is the education stage in which they choose whether to attend college or enter the labor market after graduating high school.

**(1) Education Stage:** The state variables for the students are comprised of asset $a_{IVT}$, the innate ability $h$, psychic costs $\phi$, and their parent's income $I$. They compare the expected value for entering the labor market as a high school graduate, $\mathbb{E}V^w$, with the value for enrolling in college $V_{g1}$, which is the expected lifetime utility conditional on enrolling in college, net of the psychic cost $\phi$. They choose college enrollment if the latter is greater than the former; otherwise, they enter the labor market. The decision problem is formulated as follows:

$$V_{g0}(a_{IVT}, \phi, h, I) = \max_{e \in \{0,1\}} \Bigg\{ (1-e) \cdot \mathbb{E}_{z_0} \underbrace{\left[ V^w(a_{IVT}, j = 18, z_0; e = 0, h) \right]}_{\text{Lifetime utility for high school graduates}} + e \cdot \underbrace{\left[ V_{g1}(a_{IVT}; h, I) - \phi \right]}_{\text{Value of attending college}} \Bigg\}, \tag{3}$$

where $e \in \{0, 1\}$ indicates the education choice where $e = 1$ means college enrollment and $e = 0$ does entering the labor market as a high school graduate. $V^w$ denotes the value function for workers, representing the lifetime utility of entering the labor market immediately after high school graduation. The formulations of $V^w$ and $V_{g1}$ are provided in Appendix E.. The initial draw of $z$, $z_0$, is uncertain and is according to the invariant distribution of $z$. Proceeding to college will increase their lifetime earnings, while this return from education depends on innate ability. In addition to the psychic costs, there are other two types of costs of proceeding to college. First, they have to pay tuition fees $p_{CL}$ while enrolling in college. Second, although they can supply labor and earn some income, they have to spend $\bar{t}$ fraction of their time on study, which is the opportunity costs of going to college. They can fund their consumption and tuition fees through (1) transfers from their parents $a_{IVT}$, (2) labor earnings by themselves, (3) borrowing through student loans if eligible, (4) government-provided grants if eligible.

**(2) Early Working Stage:** The household problem in this period ends up with a standard lifecycle problem in which households choose consumption-savings and leisure-labor decisions. The value function is formulated in Appendix E..

**(3) Fertility Stage:** When they reach the age of $j = J_F$, they choose how many children they have. The state variables in this stage consist of asset position $a$, productivity $z$, education $e$, and innate ability $h$. The problem is to choose the number of children $n$ that maximizes expected utility from this point onward, with the value function formulated as follows:

$$V^f(a, z, e, h) = \max_{n \in \{0,1,2,3,4\}} \left\{ V^{wf}(a, j = J_F, z; e, h, n) \right\},$$

where $V^{wf}(a, j, z; e, h, n)$ in the parenthesis represents the value function for the middle working stage with $n$ children represented below.

**(4) Middle Working Stage (with children if applicable):** The state variables in this stage consist of asset position $a$, age $j$, , productivity $z$, education $e$, innate ability $h$, and the number of children $n$. The value function $V^{wf}$, after fertility choices until their children become independent (i.e., for age $j \in \{J_F, ..., J_{IVT} - 1\}$), is formulated as follows:

$$\begin{aligned} V^{wf}(a, j, z; e, h, n) = \max_{c,l,q,a'} \{ & u(c/\Lambda(n), l) + b(n) \cdot v(q) \\ & + \left\{ \begin{array}{ll} \beta \mathbb{E}_{z'}[V^{wf}(a', j+1, z'; e, h, n)] & \text{if } j \in \{J_F, ..., J_{IVT} - 2\} \\ \beta \mathbb{E}_{z', \phi_k, h_k}[V^{IVT}(a', z'; \phi_k, h_k, e, h, n)] & \text{if } j = J_{IVT} - 1 \end{array} \right\} \\ \text{s.t.} & \\ & (1 + \tau_c)(c + nq) + a' = Y_{wf}, \\ & a' \geq -\underline{A}, z' \sim \pi(z', z), h_k \sim g_h^{h_k}, \phi_k \sim g_{e,h_k}^{\phi_k}, \end{aligned}$$

where

$$\begin{aligned} Y_{wf} \equiv\ & (1 - \tau_w) w_e \eta_{j,z,e,h} (1 - l - \kappa \cdot n) \\ & + n \cdot B + \psi + \left\{ \begin{array}{ll} (1 + (1 - \tau_a) r) a & \text{if} \quad a \geq 0, \\ (1 + r^-) a & \text{otherwise.} \end{array} \right. \end{aligned}$$

Here, $\phi_k$ and $h_k$ denote the psychic costs of college education and innate ability for their children. The household consumption is deflated by the equivalence scale $\Lambda(n)$, depending on the number of children $n$. The expenditures on children, $nq$, enter the budget constraint, and utility from children, $b(n) \cdot v(q)$, enters the objective function in the periods with children. $V^{IVT}$ represents the value function for the IVT stage at $j = J_{IVT}$, which is described in the following paragraph. Households are uncertain about their children's innate ability $h_k$ and psychic costs $\phi_k$ until the beginning of age $j = J_{IVT}$. $\tau_c$, $\tau_w$, and $\tau_a$ denote tax rates on consumption, labor income, and capital income.

**(5) Inter Vivos Transfers Stage:** At the beginning of age $j = J_{IVT}$, households decide how much to transfer to their children. This timing corresponds to when the children graduate from high school and face the college enrollment choice. Two characteristics of their children are realized at this point: their psychic costs $\phi_k$ and innate abilities $h_k$. After observing the realized characteristics, households choose the optimal amount of per-child transfer $a_{IVT}$, formulated as follows:

$$V^{IVT}(a, z; \phi_k, h_k, e, h, n) = \max_{a_{IVT} \geq 0} \Big\{ V^w(a - \tilde{a}_{IVT}, j = J_{IVT}, z; e, h) \\ + b(n) \cdot \lambda_a \cdot V_{g0}(a_{IVT}, \phi_k, h_k, I) \Big\},$$

where $\tilde{a}_{IVT} = \frac{n \cdot a_{IVT}}{1+(1-\tau_a)r}$ and the parent's state vector pins down household income $I$. They care about the lifetime utility of each child, $V_{g0}(a_{IVT}, \phi_k, h_k, I)$, based on altruism, discounted by an altruistic discount factor $\lambda_a$.

**(6) Late Working Stage:** At this stage for households aged $j \in \{J_{IVT,\dots,J_R-1}\}$, the household problem is similar to the early working stage: they make consumption-saving and leisure-labor decisions until retirement. The value function is formulated in Appendix E..

**(7) Retirement Stage:** At the beginning of age $J_R$, households are forced to retire from the labor market. After that, they spend all their time on leisure and make consumption-saving decisions. Two points differ from previous choice problems; they receive the pension benefit $p$ from the government and face uncertainty about the next period's survival. The value function is formulated in Appendix E..

## 4.4 Firm

A representative firm chooses labor and capital inputs in competitive factor markets to produce final goods. There are two types of labor inputs in this economy; the college graduates (skilled) and high school graduates (unskilled). Their total labor supply in efficiency units are represented as $L_{CL}$ and $L_{HS}$, respectively. I allow them to be imperfect substitutes by considering the aggregate labor in efficiency units, $L$, is given as:

$$L = \left[\omega_{HS} \cdot (L_{HS})^{\chi} + \omega_{CL} \cdot (L_{CL})^{\chi}\right]^{1/\chi},$$

where $\omega_{HS} \equiv 1 - \omega_{CL}$ and $\omega_{CL} \in [0, 1]$ governs the relative productivity of the skilled workers, and $\chi$ governs the elasticity of substitution between the skilled and unskilled workers. The representative firm operates with a Cobb-Douglas production function with aggregate capital $K$ and labor $L$ to produce the output $Y$:

$$Y = ZK^{\alpha}L^{1-\alpha},$$

where $Z$ represents the factor neutral productivity. Capital depreciates at the rate of $\delta$, and the firm has to incur the capital depreciation costs.

## 4.5 Government

The government raises the revenue by levying three types of taxes: consumption, labor income, and capital income taxes, where each tax rate is represented as $\tau_c$, $\tau_w$, and $\tau_a$. In addition, the government collects accidental bequests and devotes them to cover expenditures. They use this revenue to fund (1) the public pension benefits, which gives $p$ units of money to retired households each period, (2) subsidized loans for college students, (3) grants for eligible college students, where the payment per eligible student is represented by $\bar{g}(I)$ where $I$ represents household income when the student is in the education stage (i.e., their parent's age is $J_{IVT}$), (4) lump-sum transfers $\psi$ that is introduced for replicating the progressivity of labor income tax schedule in a simple way following the literature, (5) cash benefits for households with children under 17 with per-child payment of $B$, and (6) other expenditures $S$. The government budget constraint is given as follows:

$$\tau_c \cdot C + \tau_w \cdot (w_{HS}L_{HS} + w_{CL}L_{CL}) + \tau_a \cdot r \int_{\boldsymbol{x}} \mathbb{I}_{a(\boldsymbol{x})>0} a(\boldsymbol{x}) dF(\boldsymbol{x}) + Q$$
$$= p \cdot \mu_{old} + (\iota - \iota_s) \cdot K_s + G + \psi + B \cdot \mu_{j\leq 17} + S, \quad (4)$$

where $C$, $Q$, $\mu_{old}$, $\mu_{j\leq 17}$, $K_s$, and $G$ represent the total consumption, total accidental bequests, population mass of retired households, that of children under age 17, the total amount of borrowing by college students, and the total grant payments. The term $\tau_a \cdot r \int_{\boldsymbol{x}} \mathbb{I}_{a(\boldsymbol{x})>0} a(\boldsymbol{x}) dF(\boldsymbol{x})$ represents revenue from capital income taxation, where $\boldsymbol{x}$, $a(\boldsymbol{x})$, $\mathbb{I}_{a(\boldsymbol{x})>0}$, and $F(\boldsymbol{x})$ represent the household state vector, household's policy function for savings, an indicator function returning one if $a(\boldsymbol{x})$ is positive, and distribution over the state space.

### 4.6 Stationary Equilibrium

I solve the stationary equilibrium of this economy. In equilibrium, households make every choice to maximize their expected utility, the firm maximizes its profit, and the government budget is balanced. Stationarity implies that the distribution over state variables is invariant. Importantly, the age distribution is determined endogenously according to households' fertility choices. See Appendix F. for the detailed definition of equilibrium and Appendix G. for the computational algorithm for solving the equilibrium.

## 5 Calibration

To calibrate the model, I use the Japanese Panel Survey of Consumers (JPSC), a panel survey of Japanese women and their household members. It started in 1993 with a representative sample of 1,500 women aged $24 - 34$ and contains information about, for example, their income, educational background, marital status, fertility, and expenditures to detailed categories, including those on children's education. I focus on the cohort born in 1959-69, the oldest cohort of this survey, especially to compute the completed fertility and intergenerational mobility of education. I keep only married households as in previous works (e.g., Daruich and Kozlowski, 2020) because the model focuses on choices made within married households such as fertility and educational investments.[12]

[12]Hence, an agent or household in this model refers to households with two individuals. Then, "children" in this model can also be interpreted as a household unit. That is, having $n$ children can be interpreted as reproducing $n/2$ units of households. For example, if a household has two children, it means reproducing one household unit with two individuals.

Unless otherwise mentioned, targeted moments described below are computed based on the JPSC's 1959-69 cohort data.

As is standard, I follow a two-step calibration procedure. First, I set some parameters based on the literature or estimate them outside of the model. Then, the remaining parameters are calibrated internally to replicate a set of targeted moments. Hereafter, I explain the functional forms and the procedure for calibrating the relevant parameters, based on each category such as preferences and technology. Tables 11 and 12 in Appendix H. (iii) summarize the parameters determined externally and those calibrated internally. Table 5 reports the model-implied moments alongside their data counterparts.

**Preferences:** Instantaneous utility for households are given as follows:

$$u(c,l) = \frac{(c^{\mu} l^{1-\mu})^{1-\gamma}}{1-\gamma}.$$

$\mu$ is internally determined as 0.23 so that the households spend one-third of the total disposable time on market work. Instantaneous utility from expenditures on a child is given as:

$$v(q) = \lambda_q \frac{q^{1-\gamma}}{1-\gamma},$$

where $\lambda_q$ is set to 0.62 so that the annual educational expenditure per child amounts to 7% of average income at age 28. The utility must be always positive (or always negative) in models of altruism with endogenous fertility, and I set $\gamma = 0.5$ following the literature (e.g., Daruich and Kozlowski, 2020). The altruistic discount factor $\lambda_a$ is set to 1.03 so that the annual average IVT for college students amounts to 27% of average income at age 28.[13] Following Kim, Tertilt, and Yum (2023), I let the discount function of the number of children $b(n)$ be non-parametric, and assume that $b(n) = b_n$ for each $n \in \{0, 1, 2, 3, 4\}$ with $b(0) = b_0 = 0$. Those parameters are determined so that the model replicates the distribution of the completed fertility. The time for studying $\bar{t}$ is set to 0.8 so that the income share of labor earnings for college students in the model is close to 21% (SLS, 2018). I assume that $\beta = 0.98$ as in Zhou (2022).

**Financial Markets:** I set the borrowing limits $\underline{A} = 20$ million yen and $\underline{A}_s = 2.88$ million yen.[14] The borrowing wedges are set $\iota = 0.055$ and $\iota_s = 0.054$ so that the model approximates the share of working households with a negative net worth (54%) and the share of students borrowing from the government-provided student loans (44%).

**School Taste:** I assume that the psychic costs $\phi$ are given as $\phi = \psi_{CL} \cdot \exp(-\nu \cdot h) \cdot \tilde{\phi}$. First, $\psi_{CL}$ governs the scale of psychic costs and thus college enrollment rate. The second term $\exp(-\nu \cdot h)$ allows high ability students to have smaller psychic costs, as standard in the literature, and $\nu$ governs the education sorting by ability. Finally, $\tilde{\phi}$ is stochastic, and

[13] Although some parents whose children do not enroll in college make IVT in the model, the amount is negligibly small. One reason is that the marginal gains from IVT are more significant if their children attend college, as students are financially constrained, primarily because of their limited earning ability.

[14] The former is based on the Family Income and Expenditure Survey by the Ministry of Interna Affairs and Communications and the latter is based on the SLS (2018).

the distribution depends on their parent's education. Following Daruich and Kozlowski (2020), I assume that $\tilde{\phi}$ is distributed on an interval $[0,1]$ and follows the following CDF:

$$G^{\tilde{\phi}}_{e_p} = \begin{cases} \tilde{\phi}^{\omega} & \text{if } e^p = 0 \\ 1 - (1-\tilde{\phi})^{\omega} & \text{if } e^p = 1 \end{cases}$$

Here, $\omega$ governs the intergenerational transmission of school tastes. $\psi_{CL}$ is set to 20.8 so that the model approximates the college enrollment rate of 37.7%. $\nu$ is set to 1 in the benchmark, and $\omega$ is set to 1.71 so that the intergenerational transition matrix of education in the model matches the data counterpart as closely as possible. Table 4 reports the transition matrix in the model and data. $(i,j)-$th entry of the matrix indicates the probability that children acquire skill (education) $j \in \{HS, CL\}$ given that their parent's skill is $i \in \{HS, CL\}$ in the benchmark model, and values in parentheses represent the data counterparts. The table indicates that the education level is persistent across generations: children of high school graduates attend college with a probability less than 0.3, whereas children of college graduates do with a probability approximately 0.6.

| Parents/Children | High school | College |
|---|---|---|
| High school | 0.725 (0.798) | 0.275 (0.202) |
| College | 0.412 (0.423) | 0.588 (0.577) |

Table 4: Intergenerational transition matrix of education. *Note*: $(i,j)-$th entry of the matrix indicates the probability that children acquire skill (education) $j \in \{HS, CL\}$ given that their parent's skill is $i \in \{HS, CL\}$ in the benchmark model, and values in parentheses represent the data counterparts.

**Intergenerational Transmission of Ability:** The intergenerational transmission of ability is according to the following formula:

$$\log(h) = \rho_h \cdot \log(h_p) + \varepsilon_h,$$
$$\varepsilon_h \sim N(0, \sigma_h).$$

I set $\rho_h$ to 0.30 to approximate the intergenerational income elasticity (0.3) and also set $\sigma_h$ to 0.65 so that the variance of log income at age 28 in this model is close to 0.27.

**Income Process and Education Return:** The efficiency labor of an agent aged $j$, education $e$, innate ability $h$, and productivity $z$, $\eta_{j,z,e,h}$, is given as follows:

$$\log \eta_{j,z,e,h} = \log[f^e(h)] + \gamma_{j,e} + z,$$
$$f^e(h) = h + e \cdot (\alpha_{CL} h^{\beta_{CL}}),$$
$$z' = \rho^z_e z + \zeta, \quad \zeta \sim N(0, \sigma^z_e).$$

To set $\gamma_{j,e}$, I estimate the second-order polynomial of hourly wages on age using JPSC. As reported in Table 10 in Appendix H. (iii), the income gradient on age is larger for college graduates than the rest of the workers, but the degree is modest compared with the US case (e.g., Abbott, Gallipoli, Meghir, and Violante, 2019).

Following Daruich and Kozlowski (2020), I construct a series of two-year total household income (i.e., the sum of husband's and wife's labor earnings) for each couple and estimate the process for stochastic components $z$ for each education group. The estimated persistence parameters ($\rho_e^z$) for high school and college graduates are 0.98 and 0.97, and parameters governing income volatility ($\sigma_e^z$) for high school and college graduates are 0.02 and 0.03, which are similar to the process estimated in Daruich and Kozlowski (2020) using the U.S. data.[15] The function $f^e(h)$ indicates that the education return depends on innate ability $h$: individuals with higher innate abilities can obtain greater returns through college education. Following Daruich and Kozlowski (2020), $\alpha_{CL}$ and $\beta_{CL}$ are determined so that the model replicates the ratio of log wage between college graduates and the rest of the population at age $j = 28$ and the log wage variance for college graduates at age $j = 28$.

**Production:** I set the capital share $\alpha = 0.33$ and $\delta = 0.07$ following Kitao (2015). $\chi$ is set to 0.39 following Matsuda and Mazur (2022). $\omega_{CL}$ is internally determined to 0.52 so that the average wage ratio between college graduates and the rest in the model amounts to 1.36. $Z$ is determined so that the wage rate for high school graduates is normalized to one in the benchmark.

**Government:** Tax rates are set to $\tau_c = 0.1$, $\tau_w = 0.35$, and $\tau_a = 0.35$ in the benchmark. The lump-sum transfer is set to $\psi = 0.01$ to match the ratio between the variance of log net income and that of log gross income (0.6). The pension benefit $p$ is set so that the government provides ¥$160,000$ per household per month. The cash transfer $B$, which we refer to "child benefit" or "typical pro-natal transfers" hereafter, is given as ¥$10,000$ per child per month, approximating the actual payment. The other expenditure $S$ is set so that the government budget constraint is balanced in the benchmark and fixed throughout the counterfactual experiments.

**Miscellaneous:** The survival probability $\zeta_{j,j+1}$ is set based on the Vital Statistics (2019).[16] Annual college tuition fees $p_{CL}$ are set to 1.05 million yen. $\kappa$ is set to 0.044 so that they spend 13.3% of their working hours on childcare.[17]

# 6 Validation

This section validates the calibrated model, particularly its fertility behavior, which is crucial for confidence in the quantitative analysis. I demonstrate that the benchmark model successfully replicates several moments not targeted in calibration: (1) fertility differentials between college and high school graduates, (2) fertility elasticity with respect to pro-natal cash transfers, (3) cross-country estimates of fertility elasticity with respect to college subsidies, and (4) the composition of students' income among parental transfers (IVT), loans, and labor earnings.

[15] The details of the estimation procedure can be found in Appendix H..

[16] See, https://www.mhlw.go.jp/english/database/db-hw/outline/index.html.

[17] See, Kitao and Nakakuni (2024).

## 6.1 The fertility differential across education levels

More educated parents have fewer children than less educated parents. According to my JPSC sample, college graduates had a completed fertility rate of 1.92, lower than 2.12 for high school graduates, resulting in a college-to-high-school fertility ratio of 0.91.[18] In this benchmark model, the completed fertility of college graduate wives is 1.79, which is lower than the high school graduate wives', 2.28, resulting in a college-to-high-school fertility ratio of 0.80. The benchmark model captures the fertility differential in that more educated parents have fewer children than less educated parents.

Why does this fertility differential result in this model? First, the opportunity costs of having children are more significant for college graduates, making child-bearing more costly for them. Second, college graduate parents are more willing to pay for the IVT, making a child more costly. This is because the children's innate abilities and psychic costs of education, $(h_k, \phi_k)$, govern education returns and the willingness to attend college, which in turn govern the marginal gains from IVT for parents.

Appendix A. presents simple static models that highlight these mechanisms in closed forms. These two mechanisms align with the major explanations proposed by the theoretical literature on the fertility differentials across income distribution:[19] (1) having children is more costly for higher-income households, provided that having children is time-intensive, and (2) higher-income households have higher demands for child quality or an advantage in parental investments.

## 6.2 The benefit elasticity of fertility

Previous studies show that cash benefits such as the child benefit and baby bonus have a significant impact on fertility. Many of them report that the benefit elasticity of fertility, the percentage increase in fertility rate against the one percent increase in the cash transfer, is about $0.1 - 0.2$. For example, Milligan (2005) studies a reform of Quebec's baby bonus and shows that an extra 1,000 Canadian dollars benefit would increase fertility by 16.9%, which implies a benefit elasticity of 0.107. Cohen et al. (2013) uses a variation in the child subsidy payment observed around 2003 in Israel, providing a larger subsidy for third or higher births. They show that the benefit elasticity of fertility was 0.176.[20]

To examine how this model performs in this respect, I conduct the following exercise. Let $B_0$ denote the per-child cash transfers for households with children under 17 in the benchmark. I solve the household problem, holding prices, tax rate, and distribution fixed, with several levels of the per-child payment $B = B_0 \cdot (1 + x)$ for some $x \in X$,

[18] This is also observed in the National Fertility Survey (NFS) provided by the National Institute of Population and Social Security Research (IPSS), another data source where we can check the completed fertility rate by different education levels. See, https://www.ipss.go.jp/site-ad/index_english/survey-e.asp. According to the NFS (2015), college graduate wives' completed fertility has been lower than less educated ones almost every survey year since 1977. The latest survey in 2015 reports that the completed fertility of wives with a college degree was 1.89, and that of high school graduate wives was 1.98.

[19] See, for example, Jones, Schoonbroodt, and Tertilt (2010).

[20] Also, González (2013) adopts the regression discontinuity design to study the effects of Spain's reform in 2007, introducing a one-time payment of 2,500 euros (about 3,800 USD) for births, almost 4.5 times the monthly minimum wage for full-time workers. It finds a statistically significant impact on fertility, increasing conceptions by $5 - 6\%$.

where I set $X = \{0.1, 0.2, ..., 1.9, 2.0\}$, a reasonable range in the context of the expansion examined in the empirical studies. This procedure yields the implied fertility rate, and given the expansion rate $x$, I compute the corresponding benefit elasticity of fertility. Finally, I take the average of those 20 values for the implied elasticity and find that the average elasticity is 0.138 (with a standard deviation of 0.025), which is consistent with the empirical estimates.

## 6.3 The education subsidy elasticity of fertility

The model also replicates the cross-country estimate of the relationship between total fertility rates and the "generosity" of the higher education subsidy measured by the government expenditures on higher education per student relative to the country's GDP per capita. I use cross-country panel data from the World Bank,[21] collecting more than 200 countries/regions over more than 50 years. I then regress the log of total fertility rates of a country $i$ in year $t$ on the generosity of the subsidy for the same country-year, controlling for country-fixed effects, time-fixed effects, and other covariates such as GDP per capita and demographic factors. I refer to the estimated coefficient for subsidy generosity as the *education subsidy elasticity of fertility*, which quantifies the percentage increase in total fertility associated with a one percentage point rise in government spending on higher education per student relative to GDP per capita.

Based on the sample of the OECD countries, the estimated coefficient is 0.0013, which is marginally significant (at 10% level). Based on the sample of the entire set of (more than 200) countries/regions, the estimated coefficient is 0.00027, which is statistically significant at the 1% level. I simulate an introduction of a uniform subsidy using the benchmark model by providing uniform cash transfers—that amount to 1% of the GDP per capita—to all college students. This simulation leads to a 0.11% increase in total fertility, which aligns with the estimate based on the OECD countries. The details of the regression analysis are provided in Appendix H. (ii).

## 6.4 Composition of students' income

Capturing the composition of students' income—how college students finance their living expenses and tuition fees—is also important as it is critical not only to students' education choices but also to parents' IVT choices and, thereby, fertility choices. According to the Student Life Survey (2018), students' income consists of three components. The largest share, 61%, comes from parental asset transfers. Labor earnings account for 21%, while the remaining 18% is financed by student loans. Although the 21% share of labor earnings is a targeted moment, the rest is not. Nevertheless, the model captures the overall income composition well: IVT and loans account for 66% and 14% of students' income, closely matching the data counterparts.

[21] See, https://databank.worldbank.org/.

Table 5: Targeted and non-targeted moments.

| Moment | Data | Model |
|---|---|---|
| *Targeted* | | |
| *College enrollment* | | |
| College enrollment rate | 0.377 | 0.376 |
| Intergenerational mobility of education | | |
| Pr(Child=High-school \| Parent=High-school) | 0.73 | 0.80 |
| Pr(Child=College \| Parent=High-school) | 0.27 | 0.20 |
| Pr(Child=High-school \| Parent=College) | 0.41 | 0.42 |
| Pr(Child=College \| Parent=College) | 0.59 | 0.58 |
| *Income inequality & mobility* | | |
| Skill premium (college- to high-school-graduates) | 1.36 | 1.48 |
| Var log wage for college graduates at age 28 | 0.14 | 0.24 |
| Var(log disposable income)/Var(log gross income) | 0.60 | 0.68 |
| Variance of log(income) at age 28 | 0.27 | 0.24 |
| Intergenerational income elasticity | 0.30 | 0.27 |
| *Parental transfers* | | |
| Average transfer / Average income at age 28 | 0.07 | 0.07 |
| Average transfer / Average income at age 28 | 0.27 | 0.27 |
| *Fertility* | | |
| Share of one child | 0.16 | 0.15 |
| Share of two children | 0.55 | 0.61 |
| Share of three children | 0.22 | 0.24 |
| Share of four or more children | 0.02 | 0.00 |
| *College students* | | |
| Income share of labor earnings | 0.21 | 0.20 |
| Share of students using loans | 0.44 | 0.34 |
| *Labor supply and borrowing* | | |
| Work hours | 0.33 | 0.30 |
| Household share with negative net worth | 0.54 | 0.45 |
| *Non-targeted* | | |
| Fertility ratio/differential (college graduates to high school graduates) | 0.91 | 0.80 |
| College subsidy elasticity of fertility | 0.13 | 0.11 |
| Benefit elasticity of fertility | $\simeq 0.10$ | 0.138 |
| Students' income composition (IVT:Loans:Labor earnings) | (0.61 : 0.18 : 0.21) | (0.66 : 0.14 : 0.20) |

# 7 Roles of an Income-tested College Subsidy

I introduce an income-tested college subsidy—which approximates the one actually introduced in Japan in 2020—into the calibrated model. More specifically, I allow for an income-tested subsidy for students in low-income households, whose income is below $\bar{I}$, and students in these households receive an amount $g$ each period while they are in college. The grant function $\bar{g}(I)$ is formulated as follows:

$$\bar{g}(I) = \begin{cases} g & \text{if } I < \bar{I} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

The income threshold $\bar{I}$ in the actual policy approximately corresponds to the 15th percentile of the household income distribution, and the payment amounts to approximately two-thirds of the average college student's expenses, including both tuition fees and living expenses.[22] I set $\bar{I}$ and $g$ based on this information, considering the income distribution and students' expenditures in the benchmark model (initial steady state). I then analyze the effect of the subsidy on fertility (Section 7.1), its long-run macroeconomic implications (Section 7.2), and the transitional dynamics (Section 7.3).

## 7.1 Effects on Fertility

The main goal here is to quantify the insurance effects of the income-tested subsidy discussed in Section 3. Therefore, I first simulate the introduction of the subsidy in a partial equilibrium setup, in which prices, tax rates, and distributions are fixed as in the benchmark, and see the (direct) effect on fertility.[23]

I decompose the direct effects on fertility into two components: insurance effects and the *price effect*. The insurance effects are those discussed in Section 3. Although the illustrative model considers an income-tested subsidy with an expected value of zero, when the expected value is nonzero, as in the actual policy considered here, an additional effect arises: the price effect, which captures the reduction in the expected monetary costs (or "price") of children.

In Appendix D. (ii), I demonstrate that the direct effects can be decomposed into insurance and price effects as follows. First, I construct a grant function that provides the present discounted value of the expected college subsidy to each household at the fertility stage, based on the joint probability that its (potential) children enroll in college and that the household is eligible for the subsidy (i.e., its income will be below the income threshold). I then compute the effects of this subsidy on fertility, referred to as the price effect, which captures the impact on fertility resulting from the reduction in the expected cost of children. My decomposition formula indicates that, under certain assumptions, the insurance effect can be computed as the difference between the overall direct effect and the price effect. Tables 13 and 14 summarize the decomposition results.

[22] Although the income threshold $\bar{I}$ and payment $g$ in the actual system vary with certain characteristics of households and students, such as family structure and whether the student commutes to college from home, I abstract away from those details for simplicity.

[23] The effects on fertility in the long run, which also capture the indirect effects such as GE effect, taxation effect, and distributional effects, are reported in Appendix J.. It turns out that the direct effects explain two-thirds of the overall effects.

The main results of the decomposition are threefold. First, the insurance effects account for 62% of the direct effects on the total fertility rate (TFR) (i.e., 1.3 p.p. over a 2.1 % increase), while the price effects account for 38% (i.e., 0.8 p.p. over a 2.1% increase). This means that the insurance effect is quantitatively more important in accounting for the effect of the income-tested college subsidy on fertility. Second, the the effects on fertility are more significant for college graduates (a 4.5% increase) than high school graduates (a 0.9% increase). This is because college graduates have a higher probability of being eligible, as their children are more likely to attend college, resulting in both stronger average and insurance effects for them compared to high school graduates. Third, the relative importance of insurance effects to price effects is more significant for high school graduates: the insurance effect accounts for about half of the fertility increase for college graduates, whereas it does about 90% of that for high school graduates. This is because (1) on average, college graduates are more able to self-insure against income risk due to higher earnings, and (2) since children of high school graduates are less likely to attend college, the effect of reducing the expected cost (i.e., the price effect) is generally negligible.

| | *Average* | *Insurance* | Total |
|---|---|---|---|
| Total fertility rate ($\Delta$ %) | +0.8 | +1.3 | +2.1 |
| High school graduates ($\Delta$ %) | +0.1 | +0.8 | +0.9 |
| College graduates ($\Delta$ %) | +2.4 | +2.1 | +4.5 |

Table 6: Direct effects on fertility and decomposition results.

Although the setup of the quantitative model differs from that of a simple model presented in Section 3, the interpretation of the insurance effect still applies. First, having one more child can be, ex-post, extremely costly in utility terms in two ways after the bad income realization. First, holding the amount of IVT constant, a bad income shock can lead to an extremely low consumption level for parents if they have more children. On the contrary, holding the household consumption level constant, a bad income shock may make it infeasible for them to spend enough IVT to let their children attend college if they have more children. Both situations are costly for parents, leading to a significantly high expected marginal cost of children. If the income-tested college subsidy is available, the expected marginal costs of children in utility terms become lower because they can receive some cash after facing a bad income shock and devote it to support their children's education.[24]

## 7.2 Long-run Macroeconomic Implications

Next, I examine the long-run macroeconomic implications of the subsidy by solving the stationary equilibrium with the grant function $\bar{g}(I)$. The government adjusts the labor

[24] Appendix N. examines other policies, including those insuring against income risks, such as (fertility-dependent) wage insurance, in an expenditure-neutral manner. I show that although the fertility-dependent wage insurance has a positive effect on fertility, the effect is smaller than the insurance effects of income-tested college subsidy. This is because the income-tested college subsidy provides partial insurance not only against income risk but also against another source of uncertainty: uncertainty regarding children's characteristics (i.e., psychic costs and innate abilities) and the associated uncertainty regarding the willingness to pay for children's education.

income tax rate $\tau_w$ to balance its budget. The main result here is that the effects of introducing the subsidy—such as a higher college enrollment rate, lower skill premium and income inequality, and increases in output and welfare—are quantitatively more significant if we take into account endogenous fertility.

I highlight the roles of endogenous fertility by solve the *exogenous fertility* version of the model, where the household decision rule (policy function) for fertility, $\bar{n}(a, z, e, h)$, are fixed as in the benchmark. Note that this exogenous fertility setup is common in the macroeconomic literature on college subsidies (e.g., Krueger and Ludwig, 2016; Matsuda and Mazur, 2022). Comparing the results under exogenous fertility with those under endogenous fertility reveals the role of fertility choices in evaluating the macroeconomic implications of a college subsidy. Table 7 reports the main results under both endogenous and exogenous fertility.

| | Fertility setup | |
|---|---|---|
| | Endogenous | Exogenous |
| College enrollment ($\Delta$ p.p.) | +4.0 | +3.5 |
| Skill premium ($\Delta$ %) | −8.3 | −5.8 |
| Standard deviation of log wage ($\Delta$ %) | −1.3 | −1.1 |
| Output per capita ($\Delta$ %) | +0.7 | +0.5 |
| Labor income tax ($\Delta$ p.p.) | +0.02 | +0.13 |
| Welfare (%) | +5.1 | +3.5 |
| Marginal Value of Public Funds (Willingness to pay / Net costs of policy) | 302.94 | 17.06 |

Table 7: Long-run macroeconomic implications (endogenous vs. exogenous fertility). *Note*: For "Standard deviation of log wage," wages for an agent with $(j, z, e, h)$ is captured as $w_e \cdot \eta_{j,z,e,h}$. Welfare gains are represented in terms of consumption equivalence for newborn agents under the veil of ignorance.

First, the policy introduction results in a 3.5 p.p. increase in the college enrollment rate under exogenous fertility, which is lower than a 4.0 p.p. increase under endogenous fertility. Under both exogenous and endogenous fertility, the grants enable some children who would otherwise be unable to afford college to enroll. Also, an increase in the college enrollment rate is amplified in the long run through a distributional effect: the composition of college graduates in younger generations increases, and when they reach parenthood, their children are more likely to enroll in college due to intergenerational linkages via assets, abilities, and tastes.[25]

Under endogenous fertility, the college enrollment rate can further increase through an additional channel. As discussed in Section 7.1, the subsidy increases fertility particularly among college graduates, and their children are more likely to attend college due to intergenerational transmission of abilities and tastes. When these children become parents themselves, their own children, if they have any, are also likely to inherit similar traits, favoring college enrollment. In other words, fertility responses amplify the distributional effect on college enrollment. Through this fertility channel, the long-run share of college

[25] A detailed decomposition exercise can be found in Appendix J..

graduates increases further, explaining why the long-run college enrollment rate is higher under endogenous fertility.

A higher college enrollment rate implies a greater supply of college graduates in the labor market, leading to a reduction in the skill premium. The decline in the skill premium under endogenous fertility (−8.3%) is more significant than under exogenous fertility (−5.8%) due to higher college enrollment rates under endogenous fertility. When considering another proxy for income inequality, such as the standard deviation of log wages, the reduction in income inequality is also more significant under endogenous fertility (−1.3%) than under exogenous fertility (−1.1%), partly due to the lower skill premium.

The output per capita increases more under endogenous fertility (+0.7%) than under exogenous fertility (+0.5%), due to greater supply of college-graduate workers. The greater output implies the greater tax bases, leading to a lower tax rate in equilibrium under endogenous fertility (35.02%) than under exogenous fertility (35.13%). The welfare gain—measured as consumption equivalence for new born agents under the veil of ignorance—is greater under endogenous fertility (+5.1%) than exogenous fertility (+3.5%) because of those channels (i.e., a lower tax, lower inequality, and a higher expected lifetime income).[26]

I also compute the average Marginal Value of Public Funds (MVPF), defined as the average willingness to pay for a policy divided by the aggregate (net) costs of the policy. The MVPF under endogenous fertility (302.94) is significantly greater than under exogenous fertility (17.06) because the policy is almost self-financing under endogenous fertility.[27]

In Appendix K., I examine the expansion of the policy by increasing the income-threshold $\bar{I}$. The main results are threefold. First, the expansion of the policy further increases college graduates' fertility, and this fertility response contributes to a higher college enrollment rate via intergenerational linkages. Second, however, this expansion reduces the fertility of high school graduates because it significantly increases the intergenerational mobility of education and results in a "quantity-quality trade-off" of children. Third, the positive effects on output during expansion diminish and eventually turn negative as the threshold $\bar{I}$ becomes sufficiently high, since broader coverage significantly crowds out labor supply and savings among households with children.

## 7.3 Transitional Dynamics

Lastly, I investigate the transitional dynamics of the economy upon the introduction of the income-tested subsidy. The introduction occurs unexpectedly for households in the initial equilibrium (benchmark economy). Households have perfect foresight about future prices and tax rates and make subsequent choices to maximize utility. For computational reasons, I assume that the government collects the lump-sum taxes to balance the budget

[26] In Appendix I., I formulate this welfare measure and discuss theoretical and conceptual difficulties on welfare analysis with models with endogenous fertility.

[27] Hendren and Sprung-Keyser (2020) examine the MVPFs for 133 historical policy changes over the past half-century in the U.S. and find that the MVPFs for college education subsidies were infinite in many cases, as they were self-financing. Krueger et al. (2024) also show that education policies, such as making college free and improving funding for public schooling, are more than self-financing using a GE-OLG model.

during the transition and that the labor income tax rate immediately reaches its final steady-state value, $\tau_w = 0.3502$.[28]

The takeaway here is that it takes a long time for the macroeconomic effects to accrue, and that part of the reason is fertility responses are taken into account. For example, the left-panel of Fig 2 represents the transition path of college enrollment rates. College enrollment rates (within cohorts) increase upon the introduction but do not immediately reach the new equilibrium value and gradually converge to it. This is because a part of the long-run increase in college enrollment rate is explained by the distributional (composition) effects: students who benefit from the policy become parents in the future, whose children will be more likely to attend college due to intergenerational persistence of assets, abilities, and tastes. Also, the fertility responses strengthen the distributional effects: college graduates increase fertility, and those "marginal" children born are more likely to be college graduates, whose children will also be more likely to attend college. The roles of fertility margin can be observed in the difference between the transition path under endogenous fertility (a blue solid line) and that under exogenous fertility (a red dashed line). Both paths for college enrollment rates are nearly identical for the first 18 years after the policy introduction but begin to diverge thereafter. This divergence corresponds to the period when the first generation born after the policy starts making college enrollment choices.

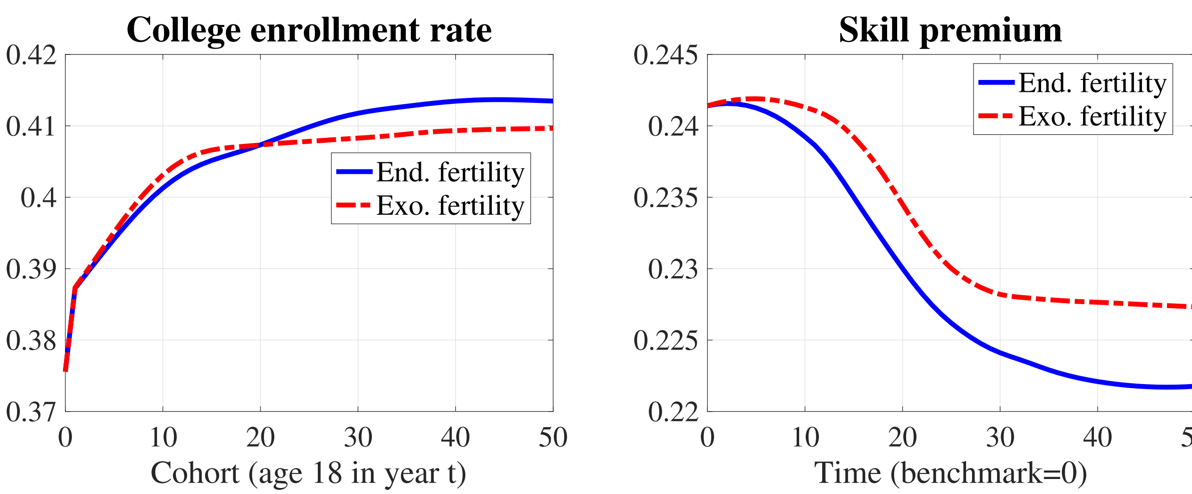


Fig 2: Transitional dynamics of college enrollment rates and skill premium. *Note*: "End. fertility" and "Exo. fertility" represent transition paths under endogenous fertility and exogenous fertility.

The right-panel of Fig 2 shows the transitional path of the skill premium. It decreases over time as college enrollment rates increase. Under exogenous fertility, the skill premium reaches its final value 30 periods (60 years) after the policy introduction, whereas under endogenous fertility, it takes more than 40 periods (80 years). This is because college enrollment rates take longer to reach their final steady-state level under endogenous fertility.[29]

[28]Some previous studies also adopt this strategy for computational reasons (e.g., Daruich, 2022).

[29]Transitional paths for other variables, such as taxes, output, and welfare, are presented in Appendix O..

# 8 Concluding Remarks

This paper examines the macroeconomic implications of income-tested college subsidies, which are prevalent across countries, by constructing a lifecycle Aiyagari model with fertility and college enrollment choices. The main contributions of this paper are: (1) to analytically and quantitatively show that income-tested subsidies provide insurance for fertility choices under income uncertainty and (2) to show that the resulting fertility responses amplify the macroeconomic effects of the policy.

Although the model is applied to Japanese data, the underlying mechanisms, highlighted through both analytical and quantitative models, are not unique to Japan but offer insights applicable to other countries as well. Furthermore, while this study focuses on college education subsidies, the fertility insurance mechanisms also apply to income-tested subsidies at earlier education levels and to a broader set of child-related transfers. Exploring these broader policies and their macroeconomic implications would require extending the current model by incorporating human capital accumulation before college.[30] This paper also abstracts competition for college admissions, which can also be important to consider when evaluating broader education policies, such as education taxes and changing enrollment capacities.[31] Incorporating these factors into the current model and exploring broader child-related and education policies are left for future research.

[30] Examples of recent works in this respect include Lee and Seshadri (2019); Daruich (2022); Yum (2023); Moschini and Tran-Xuan (2023); Krueger, Ludwig, and Popova (2024).

[31] See, for example, Kim, Tertilt, and Yum (2023); Kang (2024); Gu and Zhang (2024).

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# A. Illustrative Examples for Fertility Differentials

## A. (i) The Price of Time Theory

Following Jones, Schoonbroodt, and Tertilt (2010), I consider the following problem:

$$\max_{c,n} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \theta \frac{n^{1-\sigma}}{1-\sigma} \right\}$$

s.t.

$$c = w \cdot (1 - n \cdot \chi)$$

Here, $c$, $n$, $w$, and $\chi$ denote consumption, fertility, an exogenous wage, and time costs of children. The optimal fertility decision $n^*$ is given as:

$$n^* = \frac{1}{w^{\frac{1-\sigma}{\sigma}} \cdot (\chi/\theta)^{\frac{1}{\theta}} + \chi},$$

meaning that the negative income-fertility relationship holds if $\sigma > 1$ (i.e., the substitutability between consumption and children are high enough). Because having children is time-consuming, its (opportunity) cost is higher for high-income households. At the same time, high-income households, by definition, have higher incomes and afford to have more children. If the substitutability between consumption and children is sufficiently high, the substitution effect dominates the income effect, implying the negative income-fertility relationship.

## A. (ii) The Quantity-Quality Trade-off

Following De La Croix and Doepke (2003), I consider the following problem:

$$\max_{c,n,e} \{\ln(c) + \theta \ln(nq)\}$$

s.t.

$$c + b \cdot n \cdot e = w \cdot (1 - n \cdot \chi),$$
$$q = (\eta + e)^{\gamma},$$
$$c, n > 0, e \geq 0.$$

Here, the choice variables $c$, $n$, and $e$ represent the consumption, fertility, and per-child education investment. The investment $e$ is transformed into the "quality" of children, $q$, according to a innate ability production function $q = (\eta + e)^{\gamma}$. The parameter $\gamma \in (0, 1)$ governs the marginal gains of investment, and $\eta > 0$ implies that the quality of children takes a positive value even if parents make no investments. Parameters $\chi$ and $b$ represent the time cost of having a child and a unit (monetary) cost of investment.

The optimal solutions for the fertility and education investment are given as

$$e^* = \frac{\gamma w \chi / b - \eta}{1-\gamma}, \tag{6}$$

$$n^* = \frac{1}{1+\theta} \cdot \frac{1-\gamma}{\chi - \eta b / w}. \tag{7}$$

There are two critical observations in the optimal choices (6) and (7): (1) the optimal number of children is decreasing in wage level $w$, and (2) the optimal number of children is decreasing in the investment efficiency $\gamma$ whereas the optimal investment is increasing in $\gamma$. These two observations capture the main theoretical explanations for the negative relationship between income and fertility: (1) having children is more costly for higher-income households, provided that having children is time-intensive, and (2) higher-income households have higher demands for child quality or an advantage in parental investments.[32]

My full quantitative model captures these two channels, contributing to replicating the fertility differentials between college and high school graduates observed in the data. First, the opportunity costs of having children are more significant for college graduates, making child-bearing more costly for them. This is because having a child requires parents to incur a fixed amount of time, and college graduates' wage rates are higher than those of high school graduates. Second, college graduate parents are more willing to pay for the IVT, making a child more costly. This is because the children's innate ability and psychic costs of education, $(h_k, \phi_k)$, govern education returns and the willingness to attend college, which in turn governs the marginal gains from IVT for parents. In particular, the psychic costs of children correlate with those of parents, so college graduate parents derive higher utility from the IVTs than high school graduate parents.

# B. Fertility Choices with Income Risk

This section illustrates a simple model of fertility choices under income uncertainty and shows that the higher income uncertainty leads to lower fertility. The following setup can be considered a special case of Section 3, in which no subsidy is provided and the income distribution is uniform.

**Setup:** I consider a household problem where a household chooses the number of children, $n$, and consumption, $c$, to maximize its expected utility while facing income uncertainty. Households first choose the number of children, $n$, before the realization of the income level; the income level can take on either $y - \varepsilon$ or $y + \varepsilon$, with $\varepsilon \in (0, y)$ for some $y > 0$, with equal probabilities. Let $q$ be the unit cost of having children,[33] and for simplicity, the number of children chosen must be positive. After the income level is realized, they choose the consumption level, $c$. The household problem is formulated as

[32] For more details, see Jones et al. (2010).

[33] $q$ takes a positive value and is assumed to be lower than the worst income level, $y - \varepsilon$, to ensure positive consumption and fertility.

follows:

$$\max_{n>0} \ \mathbb{E}_y \left\{U(c) + v(n)\right\}$$
$$\text{s.t.}$$
$$c + q \cdot n = \begin{cases} y + \varepsilon & w.p.\ 1/2 \\ y - \varepsilon & w.p.\ 1/2 \end{cases}$$

Now, based on standard assumptions on utility functions, namely that (1) $v''(n) < 0$ and (2) $U'''(c) > 0$, it holds that

$$\frac{dn}{d\varepsilon} < 0, \tag{8}$$

meaning that the higher income uncertainty leads to lower fertility.

**Derivation:** I first take the first-order condition with respect to $n$ as follows:

$$\begin{aligned} q^{-1} \cdot v'(n) &= \mathbb{E}U'(c) \\ &= \frac{1}{2}[U'(y + \varepsilon - n \cdot q) + U'(y - \varepsilon - n \cdot q)] \end{aligned} \tag{9}$$

To capture the relationship between $n$ and $\varepsilon$, I then express $n$ as a function of $\varepsilon$ and take the first-order derivatives of the both sides of equation (9) with respect to $\varepsilon$ as follows:

$$\begin{aligned} 2q^{-1}v''(n(\varepsilon))\frac{dn}{d\varepsilon} &= U''(y + \varepsilon - q \cdot n(\varepsilon)) - U''(y - \varepsilon - q \cdot n(\varepsilon)) \\ &\quad - q \cdot \frac{dn}{d\varepsilon}[U''(y + \varepsilon - q \cdot n(\varepsilon)) + U''(y - \varepsilon - q \cdot n(\varepsilon))] \\ \Leftrightarrow 2q^{-1}v''(n(\varepsilon))\frac{dn}{d\varepsilon} &+ q \cdot \frac{dn}{d\varepsilon}[U''(y + \varepsilon - q \cdot n(\varepsilon)) + U''(y - \varepsilon - q \cdot n(\varepsilon))] \\ &= U''(y + \varepsilon - q \cdot n(\varepsilon)) - U''(y - \varepsilon - q \cdot n(\varepsilon)). \end{aligned}$$

Finally, we have the stated result (8) as follows:

$$\frac{dn}{d\varepsilon} = \frac{\overbrace{U''(y + \varepsilon - q \cdot n(\varepsilon)) - U''(y - \varepsilon - q \cdot n(\varepsilon))}^{>0 \ (\because\ U'''>0)}}{\underbrace{2q^{-1}v''(n(\varepsilon)) + q \cdot [U''(y + \varepsilon - q \cdot n(\varepsilon)) + U''(y - \varepsilon - q \cdot n(\varepsilon))])}_{<0 \ (\because\ v''<0\ \&\ U''<0)}} < 0$$

# C. Education Costs and Fertility Choices in Japan

Japan is a leading country in demographic aging,[34] leading to the shrinking labor force, output, and tax base, while the public expenditures on social security benefits are in-

[34] Japan's current fertility rate has been around 1.3, far below the population replacement level. In addition, the old-age dependency ratio is more than 50% and is projected to reach 80% in 2050, which is by far the highest among the OECD countries (See, https://data.oecd.org/).

creasing. As a countermeasure against this demographic issue, the government introduced grants for college students in 2020. Two key underlying premises are: (1) it will increase the "quality" of the labor force in the long run, and (2) it will increase the fertility rate, which increase the "quantity" of the labor force in the long run. This pro-natal motive is explicitly described in an act for introducing the grants.[35]

> The aim is ... to foster an environment where people can bear and raise their children with a sense of ease by alleviating the economic burden associated with higher education, thereby contributing to addressing the rapid decline in the birthrate in our country.
> (Act on Support for Higher Education Studies, enacted on May 17, 2019)

Although that expectation for education policies as a pro-natal policy is unconventional, there are facts suggesting that the financial costs for parents to support their children's college enrollment are a significant barrier to fertility in Japan. I first list the five facts below and then elaborate on each one by one:

1. Japan is one of the least in subsidizing tertiary education.
2. Average parental transfers to college students amount to 65% of the average total education expenditures up to high school graduation.
3. About 80% of parents desire a college education for their children.
4. Couples stop at fewer children than their ideal number primarily due to financial costs.
5. Children's college enrollment is associated with a higher likelihood of parents stopping at fewer children due to the financial reason.

**Fact 1.** *Japan is one of the least in subsidizing tertiary education.*

Japan is one of the least in subsidizing tertiary education, which roughly corresponds to four-year college education in Japan, given that the enrollment rates for tertiary education other than the four-year college are significantly lower than four-year college enrollment rate.[36]

To see this, I use the cross-country data provided by the OECD[37] and define the *subsidization rate* for a specific education category $e$ as follows:

$$s_e = \frac{y_e^{pub}}{y_e^{pri} + y_e^{pub}},$$

[35] See, https://www.mext.go.jp/a_menu/koutou/hutankeigen/detail/__icsFiles/afieldfile/2019/05/17/1417025_02_1.pdf (available only in Japanese).

[36] According to the Basic School Survey by the MEXT, the enrollment rate for some college was 4% in 2022 and is declining over the past thirty years.

[37] I use 2018's data given that it is the latest year in which data on a significant number of countries is available.

where $y_e^{pri}$ and $y_e^{pub}$ represent the private and public spending on education $e$, both represented as a share of the GDP. Public spending includes expenditures on educational institutions and educational-related subsidies for households or students. Private spending refers to expenditures financed by households and other private entities. Private spending includes expenditures on school but excludes those outside educational institutions (e.g., textbooks purchased by households, private tutoring, and student living costs).[38] In other words, the subsidization rate indicates what fraction of (potential) school-related costs are funded by the government.

As Table 3 shows, Japan subsidizes more than 90% of the costs for primary-to-secondary education, which is higher than the OECD average. On the contrary, the subsidization rate for tertiary education was only 32%, which is the second lowest among OECD countries and less than half of the OECD average of 70%. This fact shows plenty of room for increasing subsidies for college students, which also has driven recent policy discussions on introducing and expanding education subsidies for college students.

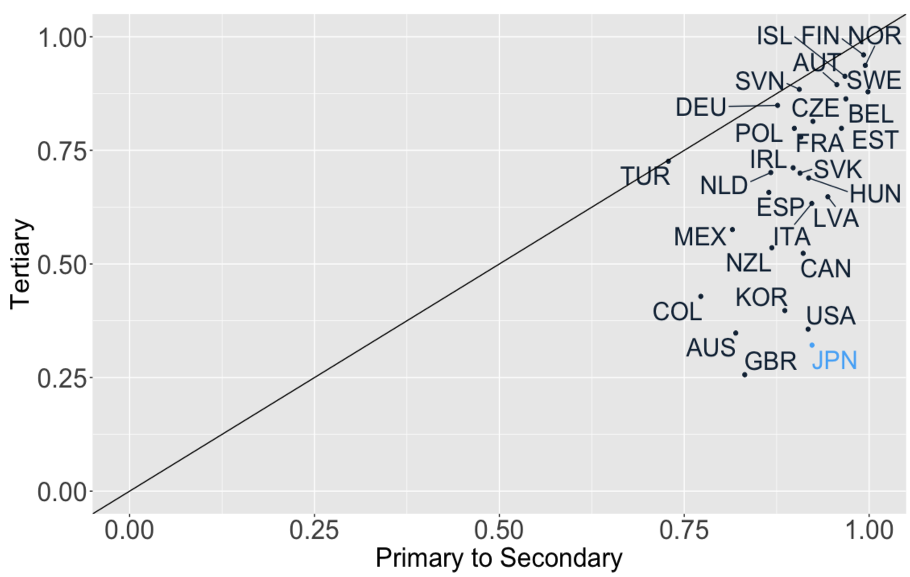


Fig 3: Subsidization rate for each education category (OECD, 2018).

Fig 4 plots the ratio of per-student government expenditures on tertiary education to per-capita GDP. On average, the government expenditures for each student enrolling in tertiary education amount to 33% of per-capita GDP, but the Japanese government only does to 21%.

[38]For more details, see the OECD data (https://data.oecd.org/).

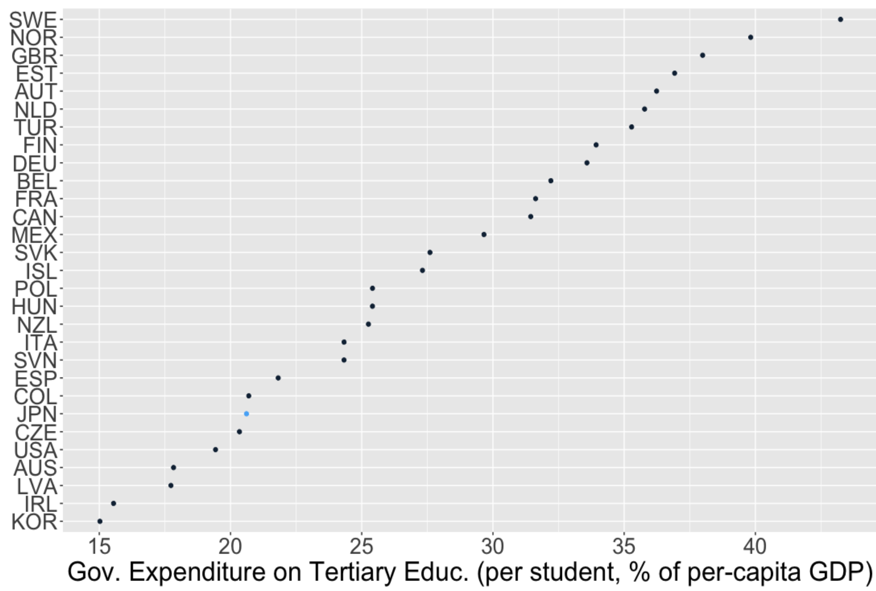


Fig 4: Per-student government spending on tertiary education to per-capita GDP (OECD, 2016-17).

**Fact 2.** *Average parental transfers to college students amount to 65% of the average total education expenditures up to high school graduation.*

I use two data sets: (1) the Survey of Children's Learning Expenses (SCLE, 2021) and (2) the Student Life Survey (SLS, 2018), both cross-sectional household surveys and conducted by the Ministry of Education, Culture, Sports, Science and Technology (MEXT).[39] The SCLE covers more than 53,000 students from preschool to high school and reports the per-student average education expenditure for each expenditure category and education stage (i.e., preschool, elementary, junior high, and high school). The expenditure category includes not only school-related ones (e.g., tuition fees and textbooks) but also extracurricular activities (e.g., cram school, music, arts, and sports). In addition to that, I use the SLS for parents' expenditures when their children enroll in college.[40]

[39]See, https://www.mext.go.jp/b_menu/toukei/chousa03/gakushuuhi/1268091.htm for the SCLE and https://www.jasso.go.jp/statistics/gakusei_chosa/__icsFiles/afieldfile/2021/03/09/data18_all.pdf (in Japanese) for the SLS.

[40]In 2018, the SLS correct answers from 43,394 students attending tertiary education, including college, some college, and graduate school.

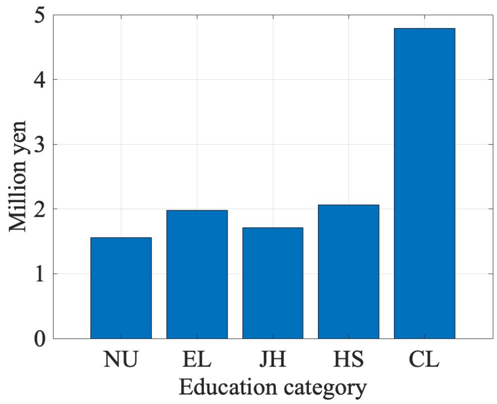


Fig 5: Average education expenditures.

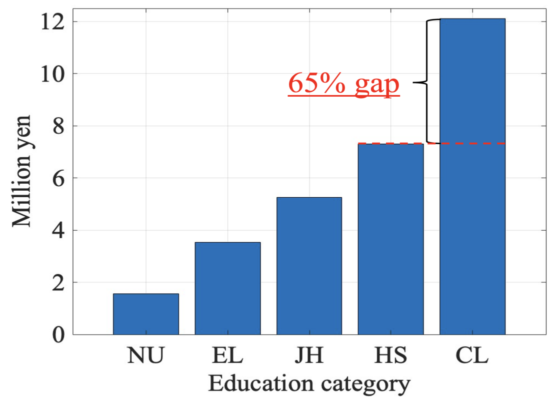


Fig 6: Cumulative education-expenditures.

Fig 5 shows the average per-child expenditure for each education category until completion. "NU," "EL," "JH," "HS," and "CL" each stand for nursery school or preschool, elementary school, junior high school, high school, and college, only consisting of a four-year college. All expenditures are conditional on enrollment in the education stage. Fig 6 represents the cumulative education expenditures, showing that the expenditure jumps up when children attend college. Given that the high school graduation rate is approximately 100% in Japan, the figure tells us that raising one child on average takes the education costs of 7.31 million yen, which amounts to 4.5% of the individuals' average lifetime labor earnings.[41] If their children attend college, they have to spend another 4.8 million yen, meaning there is about a 65% increase in education costs if parents send their children to college. This expensiveness of college education can, at least partly, be attributed to the fact that Japan is one of the least in subsidizing tertiary education while subsidizing a sizable portion of schooling costs up to secondary education. These observations establish Fact 2: A significant financial cost gap exists between those who have children enrolled in college and those who do not.

**Fact 3.** *About 80% of parents desire a college education for their children.*

We return to the NFS (2015), which asked respondents about the desired education level for their children. Fig 7 summarizes the results by wife's age. Here, "SC" stands for some college. The figure shows that approximately 75% of the parents desire a college education for their children, which is significantly higher than the college enrollment rate, which was 56.6% in 2022. This observation suggests that, although college enrollment in Japan has been far below 100%, education costs for a college education can be relevant not only for a specific part of the population because many parents would like to send their children to college.

[41]I construct a proxy of the average individual's lifetime earnings based on the 2022 Basic Survey on Wage Structure (BSWS) by the Ministry of Health, Labour, and Welfare (MHLW). First, I compute the average monthly earnings of ordinary workers for each age unconditional on any other characteristics such as sex and education. Then, I sum up the average earnings for each age, which amounts to about 160 million yen, and regard it as a proxy of the individuals' average lifetime labor earnings.

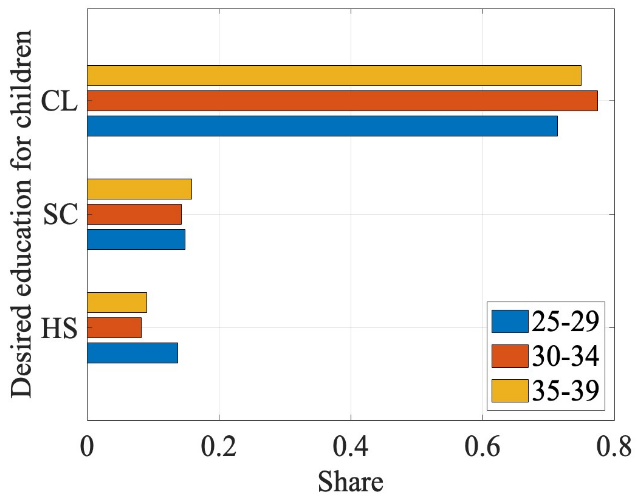


Fig 7: Wives' intention for children's education attainment.

**Fact 4.** *Couples stop at fewer children than their ideal number primarily due to financial costs.*

This fact is drawn from the National Fertility Survey (NFS), which is provided by the National Institute of Population and Social Security Research (IPSS).[42] The NFS is a cross-sectional household survey that asks respondents about their preferences or intentions regarding fertility, marriage, child-raising, and education, as well as their basic information, including their education, age, and income. It is conducted nearly every five years, and the latest survey available is in 2015, which collected 5,334 couples in which the wife is aged 18 to 49. Hereafter, I present the results focusing on married couples in which the wife is aged 25 to 39 years, leaving 2,420 couples.[43] According to the NFS, a non-negligible gap exists between the ideal and planned numbers of children. In the 2015 survey, the ideal number of children for wives aged 25 to 39 was on average 2.38, whereas the planned number was 2.16. Fig 8 represents the distribution of the ideal and planned numbers of children, where blue (red) bars indicate the share of wives who desire (plan) to have each number of children from zero to more than five. This figure suggests that the gap originates from the downward revision of the ideal at the intensive margin. There is no significant share gap between those whose ideal number is zero and those whose planned number is zero, and a substantial fraction of wives who desire three children end up with one or two children.

[42] See, https://www.ipss.go.jp/site-ad/index_english/survey-e.asp.

[43] Its 53.1% of the sample consists of couples in which the wife is over age 40, so the sample size shrinks if we target the younger couples. I focus on wives under age 39 because here I am interested in the fertility intention of those in the stage of fertility decision. Note that the cohort fertility rate is stable after age 40 for any cohort in Japan. See, for example, p7 of https://www.mhlw.go.jp/toukei/saikin/hw/jinkou/tokusyu/syussyo07/dl/gaikyou.pdf (in Japanese). Excluding those aged 18 to 24 does not affect the result significantly because they consist only of 1.5% of the observations for wives aged 18 to 49.

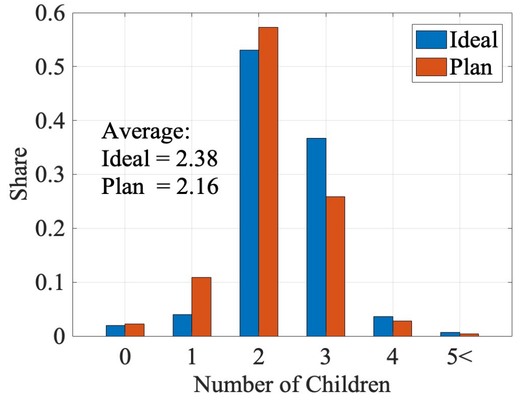


Fig 8: Distribution of ideal and planned numbers of children.

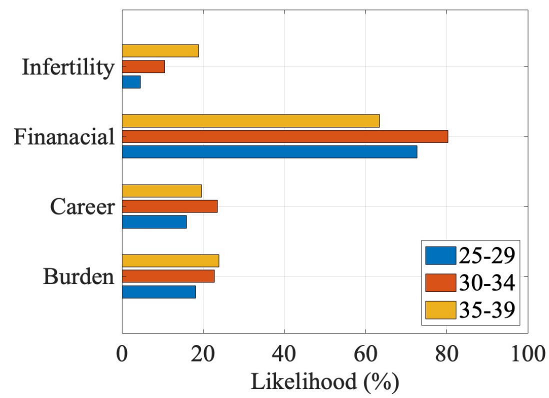


Fig 9: Reasons for the gap between the ideal and planned number of children.

Why do people abandon their ideal number of children? The NFS asks them to pick their reasons among several options, and the result indicates that they are most likely to choose the financial reason: "raising children and education are too expensive." Fig 9 represents the likelihood[44] that wives of each age group abandon their ideal number of children for a particular reason, and "Financial" represents the financial reason. Aside from this, "Career" represents "it would interfere with my job," and "Burden" represents "I would not handle the psychological and physical burden" (arising from achieving the ideal number). On average, more than 75% of them chose the financial reason, which dominates other reasons, such as "Career" and "Burden," chosen by only 20% of them.

We observe the same even using the JPSC, which asks respondents about their preferences or intentions regarding fertility, such as "Do you want more children in the future?" If they answer "No," it then asks the reasons from fourteen options, which they can choose multiple. The main options are: (1) having and educating children takes too much financial cost; (2) I would rather spend more time with my husband or for myself; (3) I want to continue my job; and (4) I cannot expect my husband's effort in child-rearing. Fig 10 reports the likelihood for each of the four reasons, where I refer to (1), (2), (3), and (4) as *Finanial*, *Time*, *Job*, and *Husband*, respectively. This figure indicates that more than 40% of parents cease reproduction due to the financial reason, which is the most important among other reasons.

[44]I use the term "likelihood" instead of "share" because it allows respondents to choose multiple options.

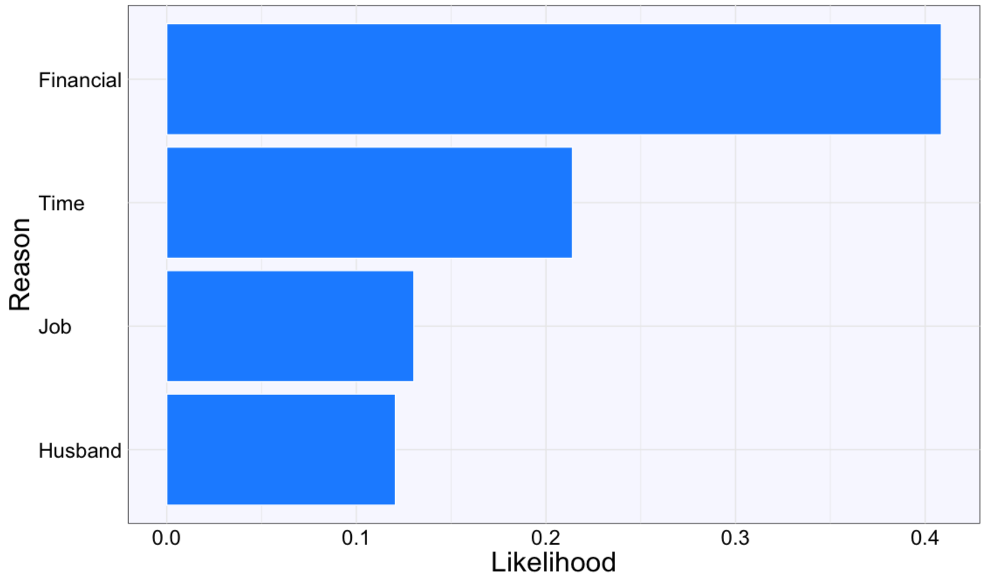


Fig 10: Likelihood for reasons why parents cease reproduction. *Note*: Each of the four reasons, *Finanial*, *Time*, *Job*, and *Husband*, represent: (1) having and educating children takes too much financial cost; (2) I would rather spend more time with my husband or for myself; (3) I want to continue my job; and (4) I cannot expect my husband's effort in child-rearing.

To sum up, there is a non-negligible gap between the ideal and planned numbers of children for couples, and a substantial fraction of them answer the reason as "raising children and education are too expensive." These results establish the first fact: couples are most likely to abandon having an ideal number of children because of financial costs.

**Fact 5.** *Children's college enrollment is associated with a higher likelihood of parents stopping at fewer children due to the financial reason.*

This fact links the above four facts and suggests that college education costs are an important determinant of fertility choices in Japan. To see this, I conduct the following regression to examine the relationship between fertility *intentions* and children's college enrollment:

$$\Pr(y_i = 1) = L(\alpha + \beta \mathit{ChildCL}_i + \boldsymbol{x}_i\boldsymbol{\theta} + \varepsilon_i),$$

Here, $y_i$ is a dummy variable taking one if a couple $i$ decides to stop having an additional child due to the financial reason. The results are summarized in Table 8.

| | (1) | (2) |
|---|---|---|
| $\mathit{ChildCL}_i$ | 0.515*** | 0.870*** |
| | (0.186) | (0.222) |
| Personal characteristics | No | Yes |
| Observations | 692 | |

Table 8: Children's college enrollment and fertility intentions of women aged 24-39. *Note:* Values in parentheses represent standard errors. *** denotes statistical significance at the 1 percent level.

The first column presents the estimates where I control only for the children's college enrollment indicator. In the second column, I add couples' characteristics, such as educational backgrounds. The first and second columns report that the log odds ratio for couples choosing the financial reason to stop having children increases by 0.515 and 0.870 points for couples whose children will enroll in college. Both are statistically significant, and the latter implies a 2.4 times higher odds ratio for stopping having an additional child due to financial reasons.

# D. More on Illustrative Model

## D. (i) Derivation of Inequality (2)

First, $cov(g(I;\varepsilon), V'(c(\varepsilon)))$ is positive because, at higher (lower) income levels, $g(I;\varepsilon) = (\mathbb{E}I - I)\cdot\varepsilon$ is lower (higher), while $V'(c(\varepsilon))$ is smaller (larger) due to the assumption that $V'' < 0$. As a result, their covariance is positive.

Next, I take the first-order partial derivative of $\mathbb{E}V'(c(\varepsilon))$ with respect to $\varepsilon$, evaluated at $\varepsilon = 0$ (i.e., an economy without subsidy), which is given as follows:

$$\begin{aligned}\left.\frac{\partial \mathbb{E}V'(c(\varepsilon))}{\partial \varepsilon}\right|_{\varepsilon=0} &= \mathbb{E}\{V''(c(0))\cdot((\mathbb{E}I - I)\cdot n(0) - a(I;0)\cdot n'(0))\} \\ &= \underbrace{\mathbb{E}V''(c(0))}_{<0}\cdot\underbrace{\mathbb{E}(\mathbb{E}I - I)\cdot n(0)}_{=0} + cov(V''(c(0)), (\mathbb{E}I - I)\cdot n(0)) \\ &- \underline{a}\cdot n'(0)\cdot \mathbb{E}V''(c(0)) \\ &= \underbrace{cov(V''(c(0)), (\mathbb{E}I - I)\cdot n(0))}_{<0} - \cdot\, \underline{a}\cdot n'(0)\cdot \underbrace{\mathbb{E}V''(c(0))}_{<0}, \end{aligned} \tag{10}$$

where the sign of $cov(V''(c(0)), (\mathbb{E}I - I)\cdot n(0))$ is negative because, with realization of higher (lower) income levels, $V''$ takes higher (lower) values because of the property $V''' > 0$, whereas $(\mathbb{E}I - I)\cdot n(0)$ takes lower (higher) values, and vice versa.

Note that the sign of the term on the right-hand side of equation (10), $n'(0)$, is of our interest. Suppose that $n'(0)$ were zero or negative, meaning that introducing an income-tested subsidy leads to a lower fertility. Then, the equations (2) and (10), coupled with $cov(g(I;\varepsilon), V'(c(\varepsilon))) > 0$ in (2), imply that $n(\varepsilon) > n(0)$ for some small $\varepsilon \in \mathcal{E}$, which is a contradiction.

## D. (ii) Decomposing Direct Effects

Instead of the subsidy function $g(I;\varepsilon) = (\mathbb{E}I - I)\cdot\varepsilon$ in the illustrative model, I consider a new function $g(I;\psi,\tau) = \psi/I + \tau$, where $\psi$ and $\tau$ are new policy parameters. Let $n(\psi,\tau)$ be optimal fertility choices given policy parameters $(\psi,\tau)$. Then, the effect of income-tested subsidy scheme, $g(I,\psi,0)$, compared with an economy without subsidy, $g(I,0,0)$, can be expressed as follows:

$$\log\left(\frac{n(\psi,0)}{n(0,0)}\right) = \underbrace{\log\left(\frac{n(0,\tau)}{n(0,0)}\right)}_{\text{=price effect}} + \underbrace{\log\left(\frac{n(\psi,0)}{n(0,\tau)}\right)}_{\text{=insurance effect}}, \tag{11}$$

where a parameter $\tau$ satisfies a condition $\mathbb{E}g(I,\psi,0) = g(I,0,\tau)$ for a given $\psi \geq 0$. This condition means that the expected value of the subsidy is neutral between the income-tested subsidy scheme $g(I,\psi,0)$ and the uniform subsidy scheme $g(I,0,\tau)$. The price effect captures the effects of reducing the expected monetary costs (or "price") of children, while the insurance effect captures the effects of smoothing marginal utilities as discussed in the illustrative model, holding the expected value of the subsidy constant.

I use (11) to decompose the direct effect observed in the full quantitative model. In practice, I construct a grant function that provides the expected value of the cash transfers to each household, given the probability of the household being eligible for the subsidy. I then compute the effects of such uniform subsidy on fertility, which is the price effect. The insurance effect is then computed as the difference between the overall direct effect and the price effect.

# E. Value Functions

**Education Stage:**

$$\begin{aligned} V_{g1}(a_{IVT};h,I) &= \max_{c,l,a'}\{u(c,l) + \beta V_{g2}(a';h,I)\}, \\ V_{g2}(a;h,I) &= \max_{c,l,a'}\{u(c,l) + \beta \mathbb{E}_{z_0}[V^w(a^s(a'), j=22, z_0; e=1, h)]\}. \end{aligned} \tag{12}$$

The budget constraints differ according to eligibility to the student loans. The budget constraints for eligible students are given as follows:

$$\begin{aligned} (1+\tau_c)c + p_{CL} + a' &= (1-\tau_w)w_{HS}(1-\bar{t}-l) + \psi + \bar{g}(I) \\ &\quad + \begin{cases} (1+(1-\tau_a)r)a & \text{if } a \geq 0 \\ (1+r^s)a & \text{otherwise} \end{cases} \\ a' &\geq -\underline{A}_s. \end{aligned} \tag{13}$$

The rest of the students cannot access to the student loans, which implies that their budget constraints are given as follows:

$$\begin{aligned} (1+\tau_c)c + p_{CL} + a' &= (1-\tau_w)w_{HS}(1-\bar{t}-l) + \psi + \bar{g}(I) + (1+(1-\tau_a)r)a, \\ a' &\geq 0. \end{aligned}$$

College students must pay tuition fees $p_{CL}$. They can fund the consumption and tuition fees through (1) transfers from their parents $a_{IVT}$, (2) borrowing through student loans if eligible, (3) government-provided grants if eligible, and (4) labor earnings by themselves. Students must spend a $\bar{t}$ fraction of their time on study while in college. Thus, they choose the time allocation between leisure and working over the disposable time $1-\bar{t}$, where the total disposable time is normalized as 1. One unit of labor supply gives college students $w_{HS}$ units of wages.[45] Following the literature, I assume that fixed payments are made for 20 years (10 periods) following college graduation and transform college loans into regular bonds according to the following formula:

$$a^s(a') = a' \times \frac{r^s}{1-(1+r^s)^{-10}} \times \frac{1-(1+r^-)^{-10}}{r^-}.$$

**Early Working Stage:** The remaining component of the value functions during the education stage, $V^w$, represents value functions in the early working stage (before fertility stage) for households aged $j \in \{J_E, \ldots, J_F - 1\}$. The state variables for this stage consists of asset $a$, age $j$, idiosyncratic component of labor productivity $z$, education level $e$, and innate ability $h$. The uncertainty in this stage is only about the next period's productivity, which is denoted by $z'$ following a Markov process $\pi_z(z', z)$. Households choose consumption, leisure, and savings given the state variables. The value function is formulated as follows:

$$\begin{aligned} V^w(a,j,z;e,h) = \max_{c,l,a'} & \{u(c,l) \\ & + \left\{ \begin{array}{ll} \beta \mathbb{E}_{z'}[V^f(a',z',e,h)] & \text{if } \ j = J_F - 1 \\ \beta \mathbb{E}_{z'}[V^w(a',j+1,z';e,h)] & \text{otherwise} \end{array} \right\} \\ \text{s.t.} & \\ & (1+\tau_c)c + a' = (1-\tau_w) w_e \eta_{j,z,e,h}(1-l) + \psi + (1+(1-\tau_a)r)a, \\ & z' \sim \pi(z',z), \ \ a' \geq -\underline{A}. \end{aligned} \tag{14}$$

Value functions $V^f$ and $V^r$ represent those for the fertility stage and retirement stage, which are formulated in Section 4.3.

**Late Working Stage:** At this stage for households aged $j \in \{J_{IVT,\ldots,J_R-1}\}$, the value function is formulated as:

$$\begin{aligned} V^w(a,j,z;e,h) = \max_{c,l,a'} & \{u(c,l) \\ & + \left\{ \begin{array}{ll} \beta [V^r(a',j+1)] & \text{if } \ j = J_R - 1 \\ \beta \mathbb{E}_{z'}[V^w(a',j+1,z';e,h)] & \text{otherwise} \end{array} \right\} \\ \text{s.t.} & \\ & (1+\tau_c)c + a' = (1-\tau_w) w_e \eta_{j,z,e,h}(1-l) + \psi + (1+(1-\tau_a)r)a, \\ & z' \sim \pi(z',z), \\ & a' \geq \left\{ \begin{array}{ll} 0 & \text{if } \ j = J_R - 1, \\ -\underline{A} & \text{otherwise.} \end{array} \right. \end{aligned} \tag{15}$$

[45] I assume that, while in college, there is no heterogeneity in labor efficiency and no uncertainty regarding the next period's productivity, and one unit of hours worked brings one unit of labor efficiency.

**Retirement Stage:** At the beginning of age $J_R$, households are forced to retire from the labor market. After that, they spend all their time on leisure and make consumption-saving decisions. Two points differ from previous choice problems; they receive the pension benefit $p$ from the government and face uncertainty about the next period's survival. The value function for the retirement stage is formulated as follows:

$$
\begin{aligned}
V^r(a, j) = \max_{c,a'} u(c, 1) + \beta \xi_{j,j+1} V^r(a', j+1)& \\
\text{s.t.}& \\
(1+\tau_c)c + a' = p + (1 + (1-\tau_a)r)a + \psi,& \\
a' \geq 0.&
\end{aligned}
$$

# F. Equilibrium Definition

Let $\boldsymbol{x}_j^e$ be an age-specific state vector for agents with education level $e \in \{HS, CL\}$ and $\mu_j^e(\boldsymbol{x}_j^e)$ be the measure of agents with state vector $\boldsymbol{x}_j^e$. Let $I_l(h, I)$ and $I_g(h, I)$ be indicator functions for loans and grants, respectively, returning 1 if students with $(h, I)$ are eligible and 0 otherwise.

Given exogenous parameters and policy rules $\{\tau_a, \tau_c, \iota, \iota_s, B, S, \psi, p, I_l(h, I), I_g(h, I)\}$, a *stationary recursive competitive equilibrium* consists of

- value functions $\{V_{g0}, V_{g1}, V_{g2}, V^w, V^f, V^{wf}, V^{IVT}, V^r\}$,
- policy functions for consumption, savings, leisure $\{c_j^e(\boldsymbol{x}_j^e), a_j^e(\boldsymbol{x}_j^e), l_j^e(\boldsymbol{x}_j^e)\}_{j=J_E}^{J}$, working hours $\{h_j^e(\boldsymbol{x}_j^e)\}_{j=J_E}^{J_R}$, fertility $\{n_{J_F}^e(\boldsymbol{x}_{J_F}^e)\}$, IVT $\{a_{J_{IVT}^e}(\boldsymbol{x}_{J_{IVT}}^e)\}$, and college enrollment $\{e_{J_E}(\boldsymbol{x}_{J_E})\}$,
- prices $(r, w_{HS}, w_{CL})$,
- labor income tax rate $\tau_w$,
- aggregate quantities $(K, L_{HS}, L_{CL})$,
- measures for households $\{\mu_j^e(\boldsymbol{x}_j^e)\}_{j=J_E}^{J}$,

such that:

1. The decision rules of students, workers, and retired households solve their problems, and $\{V_{g0}, V_{g1}, V_{g2}, V^w, V^f, V^{wf}, V^{IVT}, V^r\}$ are the associated value functions.
2. The representative firm maximizes its profit and optimally chooses capital and labor inputs:

$$r + \delta = \alpha \cdot Z \cdot \left(\frac{K}{L}\right)^{\alpha - 1}, \tag{16}$$

$$w_{HS} = \tilde{Z} \cdot \omega_{HS} \cdot L_{HS}^{\chi-1}, \tag{17}$$

$$w_{CL} = \tilde{Z} \cdot \omega_{CL} \cdot L_{CL}^{\chi-1}, \tag{18}$$

where

$$L = \left[\omega_{HS} \cdot (L_{HS})^{\chi} + \omega_{CL} \cdot (L_{CL})^{\chi}\right]^{1/\chi},$$
$$\tilde{Z} = (1-\alpha) \cdot Z \cdot \left(\frac{K}{L}\right)^{\alpha} \cdot \left[\omega_{HS} \cdot (L_{HS})^{\chi} + \omega_{CL} \cdot (L_{CL})^{\chi}\right]^{1/\chi - 1}.$$

3. The labor market for each skill $e \in \{HS, CL\}$ clears:

$$L_e = \sum_{j=J_E}^{J_R} \int_{\boldsymbol{x}_j^e} \eta_j^e(\boldsymbol{x}_j^e) \cdot h_j^e(\boldsymbol{x}_j^e)\ d\mu_j^e(\boldsymbol{x}_j^e), \tag{19}$$

where $\eta_j^e(\boldsymbol{x}_j^e)$ represents the labor efficiency of agents with a state vector $\boldsymbol{x}_j^e$.

4. The capital market clears:

$$K = \sum_{j=J_E}^{J} \sum_{e} \int_{\boldsymbol{x}_j^e} a_j^e(\boldsymbol{x}_j^e)\ d\mu_j^e(\boldsymbol{x}_j^e). \tag{20}$$

5. The government budget is balanced:

$$\tau_c \cdot C + \tau_w \cdot (L_{HS} + L_{CL}) + \tau_a \cdot K + Q = p \cdot \mu_{old} + (\iota - \iota_s) \cdot K_s + G + \psi + B \cdot \mu_{j \le 17} + S,$$

where

$$C = \sum_{j=J_E}^{J} \sum_{e} \int_{\boldsymbol{x}_j^e} c_j(\boldsymbol{x}_j^e)\ d\mu_j^e(\boldsymbol{x}_j^e),$$
$$Q = \sum_{j=J_R+1}^{J} \frac{1-\zeta_{j-1,j}}{\zeta_{j-1,j}} \sum_{e} \int_{\boldsymbol{x}_j^e} a_j^e(\boldsymbol{x}_j^e)\ d\mu_j(\boldsymbol{x}_j^e),$$
$$p \cdot \mu_{old} = \sum_{j=J_R}^{J} \sum_{e} \int_{\boldsymbol{x}_j^e} p\ d\mu_j^e(\boldsymbol{x}_j^e),$$
$$K_s = \int_{\boldsymbol{x}^s} \max\{0, -a^s(\boldsymbol{x}^e)\} \cdot I_l(h, I)\ d\boldsymbol{x}^s$$
$$G = \int_{\boldsymbol{x}^s} g(h, I) \cdot I_g(h, I)\ d\boldsymbol{x}^s$$
$$B \cdot \mu_{j \le 17} = \sum_{j=J_F}^{J_{IVT}-1} \sum_{e} \int_{\boldsymbol{x}_j^e} B \cdot n\ d\mu_j^e(\boldsymbol{x}_j^e),$$

where $\boldsymbol{x}^s$, $\mu^s(\boldsymbol{x}^s)$, and $\{a^s(\boldsymbol{x}^s)\}$ represent a state vector for college students, measure of college students, and students' policy function for saving, respectively.

6. Distributions (measures) and households' behavior are consistent.

# G. Computational Algorithm

## G. (i) Stationary Equilibrium

For any variable or distribution $x$, let $\tilde{x}$ and $\hat{x}$ represent its guessed and model-implied values. Also, let $\mu_j$ represent the distribution over state variables for age $j$. Dividing the computation process into three blocks makes it easier to understand. The first block is the outer loop, searching for equilibrium prices. The other two blocks are inner loops; one is the *optimization block*, solving household problems given prices, and another is the *distribution block*, searching for the stationary distributions given prices and policy functions obtained in the optimization block. The computational algorithm proceeds as follows:

1. Guess prices $\tilde{\boldsymbol{p}} = (\tilde{r}, \tilde{w}_{HS}, \tilde{w}_{CL})$.

2. *Optimization block*:

   - Guess the value function for agents at the beginning of age $j = 18$ ($\tilde{V}_{g0}$).
   - Given $\tilde{V}_{g0}$, solve backward from the period of IVT choice to that of the education choice, which gives the model-implied value function for $V_{g0}$, $\hat{V}_{g0}$.
   - Check if
   $$d(\tilde{V}_{g0}, \hat{V}_{g0}) < \varepsilon, \tag{21}$$
   where $d(\cdot)$ and $\varepsilon > 0$ represent an arbitrary metric function and error tolerance. If (21) is not satisfied, update $\tilde{V}_{g0}$ and follow the same procedure until convergence. The correct $V_{g0}$ pins down all value functions and policy functions given a set of prices $\tilde{\boldsymbol{p}}$.

3. *Distribution block*:

   - Guess the distribution for age $J_{IVT}$. This $\tilde{\mu}_{J_{IVT}}$ and policy functions for IVT derive the implied distribution for agents aged 18, $\hat{\mu}_{18}$. Given $\hat{\mu}_{18}$ and policy functions, compute the implied distributions for age $j = 19, ..., J_{IVT}$, and obtain $\hat{\mu}_{J_{IVT}}$.
   - Check if
   $$d(\tilde{\mu}_{J_{IVT}}, \hat{\mu}_{J_{IVT}}) < \varepsilon. \tag{22}$$
   If (22) is not satisfied, update $\tilde{\mu}_{J_{IVT}}$ and follow the same procedure until convergence.
   - After obtaing the correct distributions for age $j = 18, ..., J_{IVT}$, compute the distribution for age $j_{IVT} + 1$ onward, using those distributions and policy functions.

4. After computing value functions, policy functions, and distributions, compute the implied quantities, $\hat{L}$ and $\hat{K}$ based on (19) and (20), which gives the implied prices $\hat{\boldsymbol{p}}$ based on $\hat{L}$, $\hat{K}$, (16), (17), and (18).

5. Check if

$$d(\tilde{\boldsymbol{p}}, \hat{\boldsymbol{p}}) < \varepsilon. \tag{23}$$

If (23) is not satisfied, update $\tilde{\boldsymbol{p}}$, return to the optimization block, and follow the same procedure until convergence.

## G. (ii) Transition Dynamics

For any variable or distribution $x$, let $\tilde{x}$ and $\hat{x}$ represent its guessed and model-implied value (or distribution). Also, let $\boldsymbol{q}_t = (K_t, L_t)$ be a vector of aggregate quantities in a year $t$, and $\boldsymbol{q} = (\boldsymbol{q}_t)_{t=1,...,T}$ for some $T$. The computational algorithm for transition dynamics proceeds as follows:

1. *Preliminaries*: Set an arbitrarily large number for transition periods, $T$. Given aggregate quantities in the initial and final steady states, $\boldsymbol{q}_1$ and $\boldsymbol{q}_T$, guess a sequence of aggregate quantities, $\{\tilde{\boldsymbol{q}}_t\}_{t=1,...,T}$, where $\tilde{\boldsymbol{q}}_1 = \boldsymbol{q}_1$ and $\tilde{\boldsymbol{q}}_T = \boldsymbol{q}_T$, which pins down guesses for prices $\{\tilde{r}_t, \tilde{w}_t\}_{t=1,...,T}$. Also, guess a sequence of the labor income tax rate $\{\tilde{\tau}_{l,t}\}_{t=1,...,T}$, where $\tilde{\tau}_{l,1}$ and $\tilde{\tau}_{l,T}$ are the tax rates at the initial and final steady states.

2. *Optimization*: Given $\{\tilde{r}_t, \tilde{w}_t\}_{t=1,...,T}$ and $\{\tilde{\tau}_{l,t}\}_{t=1,...,T}$, solve the optimization problem for each cohort born before period $T$.

3. *Distribution*: Given decision rules for each cohort obtained in the above optimization block, construct the implied distributions:

   - $\tilde{\mu}_a(j;t)$: (aggregate) savings distribution across age in year $t$.
   - $\tilde{\mu}_n(j;t)$: (aggregate) labor supply distribution across working-age in year $t$.
   - $\tilde{\mu}_{j|t}$: age distribution in year $t$.

4. Given the decision rules and the implied distributions, compute the model-implied aggregate quantities, $\hat{\boldsymbol{q}}$. Also, compute the implied government revenue and expenditures, denoted by $\hat{\boldsymbol{G}}_r = (\hat{\boldsymbol{G}}_{r,t})_{t=1,...,T}$ and $\hat{\boldsymbol{G}}_e = (\hat{\boldsymbol{G}}_{e,t})_{t=1,...,T}$.

5. Check if

$$d(\tilde{\boldsymbol{q}}, \hat{\boldsymbol{q}}) < \varepsilon$$

and

$$d(\hat{\boldsymbol{G}}_r, \hat{\boldsymbol{G}}_e) < \varepsilon.$$

If the first (second) condition is not satisfied, update $\tilde{\boldsymbol{q}}$ ($\{\tilde{\tau}_{l,t}\}_{t=1,...,T}$) and follow the same procedure until convergence.

# H. More on Calibration and Validation

## H. (i) Stochastic income process

I estimate the AR(1) process for college graduates and high school graduates separately, following the procedure of Daruich and Kozlowski (2020). First, I control for the age component for each education level, $\gamma_{j,e}$, and obtain the residual $u_{i,j,e}$ for an individual $i$ aged $j$ with education level $e$, which is considered the sum of two components as follows:

$$u_{i,j,e} = m_{i,j,e} + z_{i,j,e}.$$

The measurement error $m_{i,j,e} \sim N(0, \sigma_e^m)$ is normally distributed with mean 0 and variance $\sigma_e^m$, and $z_{i,j,e}$ is a persistent shock that follows an AR(1) structure:

$$\begin{aligned} z_{i,j,e} &= \rho_e^z z_{i,j-1,e} + \epsilon_{i,j,e} \\ \epsilon_{i,j,e} &\sim N(0, \sigma_e^z). \end{aligned}$$

The initial draw for $z$, $z_{i,0,e}$, follows $N(0, \sigma_e^{z_0})$. I estimate this process for each education group using a Minimum Distance Estimator, using covariances of wage residuals at various lags for different age groups as moments.

## H. (ii) Education subsidy elasticity of fertility

The regression equation for the education subsidy elasticity of fertility, discussed in Section 6.3, is given as follows:

$$\log(\text{Fertility}_{i,t}) = \alpha + \beta \text{Spending}_{i,t} + \boldsymbol{x}_{i,t}^{'} \boldsymbol{\gamma} + \zeta_t + \eta_i + \varepsilon_{i,t},$$

where $\log(\text{Fertility}_{i,t})$ represents the log of the total fertility rate in country $i$ in year $t$, $\text{Spending}_{i,t}$ is the government spending on tertiary education (per student / GDP per capita $\times 100$) in country $i$ and year $t$, and $\boldsymbol{x}_{i,t}^{'}$ includes GDP per capita, Gini coefficient, government expenditures on other education stages, and population share of working-age population. The estimation results are summarized in Table 9.

| | (1) | (2) |
|---|---|---|
| $\text{Spending}_{i,t}$ | $2.7e-04^{***}$ | $1.3e-03^{*}$ |
| | $(7.5e-05)$ | $(7.6e-04)$ |
| | | |
| Sample | Full | OECD |
| Observations | 456 | 201 |

Table 9: Estimation results. *Note:* Values in parentheses represent standard errors. *** and * denote statistical significance at the 1 and 10 percent levels.

## H. (iii) Parameters

| | High school | College |
|---|---|---|
| Age | $+0.041$ | $+0.048$ |
| $\text{Age}^2 \times 10,000$ | $-4.551$ | $-5.364$ |
| | | |
| $\rho_e^z$ (persistence) | 0.98 | 0.97 |
| $\sigma_e^z$ (volatility) | 0.02 | 0.03 |

Table 10: Parameters governing the age profile and stochastic process for wages. *Note*: CL indicates the college graduate households where the husband or wife is a college graduate. HS represents the rest of the population.

| Parameter | Value | Description |
|---|---|---|
| $\underline{A}_s$ | 2.88 million yen | Borrowing limit for students |
| $\underline{A}$ | 20 million yen | Borrowing limit |
| $p_{CL}$ | 1.05 million yen/year | Tuition fees |
| $\kappa$ | 0.044 | Time costs |
| $\xi_{j,j+1}$ | – | Survival prob. |
| $\tau_c$ | 0.10 | Consumption tax |
| $\tau_a$ | 0.35 | Capital income tax |
| $\tau_w$ | 0.35 | Labor income tax |
| $p$ | ¥$160,000$/month | Pension benefits |
| $b$ | ¥$10,000$/month | Cash transfers |
| $\alpha$ | 0.33 | Capital share |
| $\delta$ | 0.07 | Depreciation rate |
| $\chi$ | 0.39 | Elasticity of substitution |
| $\rho_e^z$ | See Table 10 | Persistence |
| $\sigma_e^z$ | See Table 10 | Transitory |
| $\{\gamma_{j,e}\}_{j,e}$ | See Table 10 | Wage growth in age |
| $\nu$ | 1.0 | Education sorting by ability |
| $\gamma$ | 0.5 | Curvature |
| $\beta$ | 0.98 | Discount factor |

Table 11: Parameters externally determined.

| Parameter | Value | Moment | Data | Model |
|---|---|---|---|---|
| $\mu$ | 0.23 | Work hours | 0.33 | 0.30 |
| $\bar{t}$ | 0.8 | Income share of labor earnings | 0.21 | 0.20 |
| $\iota_s$ | 0.054 | Share of students using loans | 0.44 | 0.34 |
| $\iota$ | 0.055 | Household share with negative net worth | 0.54 | 0.45 |
| $\omega_{CL}$ | 0.52 | CL−HS wage ratio | 1.36 | 1.48 |
| $\psi$ | 0.01 | Var(log disposable income)/Var(log gross income) | 0.60 | 0.68 |
| $\lambda_q$ | 0.62 | Average transfer / Average income at age 28 | 0.07 | 0.07 |
| $\lambda_a$ | 1.03 | Average transfer / Average income at age 28 | 0.27 | 0.27 |
| $\omega$ | 1.71 | Intergenerational mobility of education | See Table 4 | |
| $\rho_h$ | 0.30 | Intergenerational income elasticity | 0.30 | 0.27 |
| $\sigma_h$ | 0.65 | Variance of log(income) at age 28 | 0.27 | 0.24 |
| $\psi_{CL}$ | 20.8 | College enrollment rate | 0.377 | 0.376 |
| $\alpha_{CL}$ | 0.1 | Log wage ratio (CL−HS) at age 28 | 0.34 | 0.38 |
| $\beta_{CL}$ | 0.1 | Var log wage for CL at age 28 | 0.14 | 0.24 |
| $b_1$ | 0.49 | Share of one child | 0.16 | 0.15 |
| $b_2$ | 0.53 | Share of two children | 0.55 | 0.61 |
| $b_3$ | 0.55 | Share of three children | 0.22 | 0.24 |
| $b_4$ | 0.56 | Share of four or more children | 0.02 | 0.00 |
| $Z$ | 1.99 | High school graduates' wage | 1.0 | 1.0 |

Table 12: Parameters internally determined. *Note*: "HS" and "CL" indicate high school and college graduates.

# I. On the Welfare Analysis

The welfare analysis using models with endogenous fertility is not straightforward theoretically and philosophically. One of the well-known difficulties is that the standard concept of Pareto efficiency is not applicable to models with endogenous fertility (Golosov, Jones, and Tertilt, 2007).[46] While recognizing these difficulties, this study discusses the welfare effects of the policy by computing the consumption equivalence under the veil of ignorance under the benchmark economy relative to the new steady state,[47] which is a useful standpoint for considering the potential channel through which education subsidies affect economic welfare.

To formalize the measure, let $P \in \{0, 1, 2, ...\}$ denote education subsidy schemes or policies with $P = 0$ representing the benchmark economy without grants. Next, let $V^P(\lambda)$ be the expected lifetime utility in stationary equilibrium for newborn agents with a consumption scaling parameter $\lambda$ and policy $P$:

[46] This is because models with endogenous fertility require a welfare comparison between two different sets of individuals. Golosov, Jones, and Tertilt (2007) then propose two efficiency concepts that can apply to models with endogenous fertility. For example, the $\mathcal{A}$−efficiency proposed by them focuses on individuals being alive in both economies. This concept is particularly beneficial if we examine the optimal policy as they show that the unique solution for the planning problem considering the utility of the existing agents (i.e., those who are alive just before the policy is introduced) corresponds to the $\mathcal{A}$−efficiency.

[47] The related work Zhou (2022) adopts the same welfare criteria in its steady-state analysis. In its transition analysis, Zhou (2022) also computes the average welfare of the existing households already alive before the policy changes to evaluate the $\mathcal{A}$−efficiency of the policy.

$$V^P(\lambda) = \int_{\boldsymbol{x}_1} V^P_{j=1}(\boldsymbol{x}_1; \lambda) d\mu(\boldsymbol{x}_1). \tag{24}$$

$\mu(\boldsymbol{x}_j)$ is the measure over the age-specific state space where $j \in \{1, ..., J\}$ and $V^P_{j=1}(\boldsymbol{x}_1; \lambda)$ represents the expected utility for an agent aged $j = 1$ with a state vector $\boldsymbol{x}_1$:

$$\begin{aligned} V^P_{j=1}(\boldsymbol{x}_1; \lambda) = \mathbb{E}\Bigg\{ & \sum_{j=1}^{J} \beta^{j-1} \cdot u(c_j \cdot (1+\lambda), l_j) \\ & + \sum_{j=J_F}^{J_{IVT-1}} \beta^{j-1} b(n) \cdot v(q_j) + \beta^{J_{IVT}-1} \lambda_a \cdot b(n) \cdot V_{g0}(\boldsymbol{x}_{J_{IVT}}) \Bigg\} \end{aligned}$$

Here, $\{c_j, l_j, q_j, n, a_{IVT}\}$ are optimal choices with the policy $P$ and each state $\boldsymbol{x}_j$. Finally, the consumption equivalence with a policy $P$ is given as a scalar $\lambda$ satisfying the following equation:

$$V^0(\lambda) = V^P(0).$$

That is, the consumption equivalence $\lambda$ makes the newborn agents indifferent between the new economy and the benchmark by scaling up the benchmark consumption by $\lambda$.

# J. Decomposing the Long-run Effects

The long-run increases in fertility, college enrollment rates, output, and welfare can be broken down into direct effects and indirect effects. The direct effects capture the effects of introducing the policy holding macroeconomic objects—such as prices, tax rates, and distributions—fixed and can be further broken down into the insurance effect and price effect. The indirect effects capture the effects of the policy induced by changes in factor prices (i.e., GE effects), tax rates (i.e., Taxation effects), and household distribution (i.e., Distributional effects). To isolate each effect, I conduct a decomposition exercise as follows. I first solve an equilibrium by introducing the new grant function. Then, I solve household problems by replacing one of the four objects (grant function, prices, tax rate, and household distribution) in the benchmark with that in the new equilibrium. Note that this method does not guarantee that each implied effect adds up to the overall effects because all factors except grant function are endogenous and interconnected; however, my results indicate that each effect adds up roughly to the overall effects. I further decompose the direct effects into the insurance and price effects based on the method discussed in Section D. (ii). Tables 13 and 14 summarize the decomposition results.

| | Direct | | Indirect | | | |
|---|---|---|---|---|---|---|
| | Average | Insurance | GE | Tax | Dist. | All |
| TFR (Δ %) | +0.8 | +1.3 | +1.0 | 0.0 | −1.0 | +3.1 |
| HS (Δ %) | +0.1 | +0.8 | 0.0 | 0.0 | 0.0 | −0.3 |
| CL (Δ %) | +2.4 | +2.1 | +2.4 | 0.0 | 0.0 | +10.5 |

Table 13: Decomposition results. *Note*: Columns "Direct," "GE," "Tax," and "Dist." report the results when only grant function, prices, labor income tax rate, and distribution change, respectively. A column "All" indicates the results in the benchmark and the long-run equilibrium with the grants. Rows "CL share" represent the change in college enrollment rate in p.p. and rows "HS" and "CL" represent the percentage changes in fertility rates for high school and college graduates.

**Fertility:** For the long-run increase in the TFR, vital forces are direct and GE effects. If the grant is introduced while other variables are fixed as in the benchmark, the TFR increases by 2.1% (direct effects), which corresponds to two-thirds of the overall effects.

The GE effect also plays a role in accounting for the TFR increase. If the prices are set to the long-run equilibrium levels while other objects are fixed as in the benchmark, the college graduates' fertility increases by 2.4%, and the TFR then increases by 1.0%. The key is the decline of the wage rate for college graduate workers, $w_{CL}$. In the long-run equilibrium with the grants, $w_{CL}$ decreases by 2.2%. First, the greater supply of college graduate workers depress $w_{CL}$ relative to $w_{HS}$. In addition, greater aggregate labor supply in efficiency units implies the lower marginal productivity of labor, decreasing $w_{CL}$. The lower $w_{CL}$ implies the lower opportunity costs of having children for college graduates, making some have more children. Also, for college graduate parents, the lower $w_{CL}$ reduces the (expected) utility of the IVTs, given that their children are likely to attend college, as discussed above. The lower utility of the IVTs also contributes to higher fertility because the marginal utility gains from investing in children's "quality" (by making IVTs to send a child to college) are relatively lower than those from increasing its "quantity."[48]

Lastly, the distributional effect implies a 1.0% decline in the average fertility. The introduction of the grants increases the college enrollment rate, meaning that the share of college graduates increases in the long run. Given that college graduates have fewer children than high school graduates as discussed in Section 6.1, the greater share of college graduates means a greater share of those who tend to have fewer children, which can lead to a lower average fertility.[49] However, direct and GE effects are significant and lead to the higher TFR.

[48] This fertility response of college graduates to the change in skill premium aligns with an empirical observation (Lehr, 2003).

[49] This effect is comparable with a composition effect highlighted by Zhou (2022) in understanding the effects of public education subsidies on fertility. In his model, in the long run, public education subsidies increase the share of parents with higher human capital, who tend to have fewer children.

| | Direct | GE | Tax | Dist. | All |
|---|---|---|---|---|---|
| CL share (Δ p.p.) | +2.5 | −0.1 | 0.0 | +2.2 | +4.0 |
| Output (Δ %) | −0.7 | −0.1 | −0.1 | +1.4 | +0.7 |
| Welfare (%) | +2.6 | −0.7 | −0.2 | +2.3 | +5.1 |

Table 14: Decomposition results. *Note*: Columns "Direct," "GE," "Tax," and "Dist." report the results when only grant function, prices, labor income tax rate, and distribution change, respectively. A column "All" indicates the results in the benchmark and the long-run equilibrium with the grants. Welfare gains are represented in terms of consumption equivalence.

**College Enrollment:** For the long-run increase in the college enrollment rate, critical forces are direct and distributional effects. If the grant is introduced while other variables (i.e., prices, tax rate, and distribution) are fixed as in the benchmark, the college enrollment rate increases by 2.5 p.p., more than half of the overall increase. These direct effects capture the effects of relaxing financial constraints on college enrollment decisions. GE effects put downward pressure on college enrollment because the lower college premium reduces the incentive for college enrollment.

Note that these direct effects do not explain the overall effects, so we need other forces accounting for the increase in college enrollment rate; that turns out to be the distributional effects. The implied changes in the distribution in the long run, holding grant function, prices, and tax rate fixed as in the benchmark, lead to a 2.2 p.p. higher college enrollment rate. The most relevant factor is the change in the skill distribution of parents (i.e., the share of college graduates). The increase in the share of college graduates implies a higher share of those more likely to have children enrolling in college due to the intergenerational persistence of education. Then, the distributional change amplifies the short-run increase in college enrollment rate facilitated by the direct effects.

**Output:** The distributional effect is critical in accounting for the output increase. In the long run, the working-age population increases due to a higher equilibrium TFR, and the share of skilled workers increases. These forces increase the labor supply in efficiency units, increasing output in the long run.

In contrast, the direct and GE effects put downward pressure on output. As indicated in Table 14, the GE effects lower the college enrollment rate due to the reduced skill premium. The resulting lower share of skilled workers leads to a lower output.

Next, recall that the direct effect can be considered the short-run effect of the introduction, where other macroeconomic variables, such as prices, tax rates, skill distribution, and age distribution, are fixed as in the benchmark. In the short run, aggregate savings and labor supply decrease due to higher fertility; having a child requires parents to spend a fraction of their time and some additional money, reducing their working hours and savings. Thus, the aggregate output decreases in the short run.

**Welfare:** The direct effect explains more than half of the welfare gains. Importantly, introducing the grants makes agents attend college with smaller costs and enables someone who could not attend college in the benchmark to do that, bringing them a higher lifetime income. The rest of the gains are explained by the distributional effect. The education subsidy increases the share of college graduates in the long run, and the lifetime utility for college graduates is, in principle, greater than that for high school graduates, particularly because of their higher income due to the college education returns. Thus, this distributional change increases the expected lifetime utility. Note that the direct effect is about the change in $V^P_{j=1}$ in equation (24), while the distributional effect is about the change in $\mu$ there.

# K. Expansion

In this section, I consider the effects of raising the income threshold $\bar{I}$ so that students in households of broader income classes can be eligible. Recall that I set the threshold $\bar{I}$ so that the existing grants target the students in households at the bottom 15 % of the income distribution. In this experiment, I increase $\bar{I}$ so that it corresponds to the 40, 50, and 60 percentiles of the income distribution and solve the stationary equilibrium in each case. For students in the household at the bottom 15 %, the payment is still given by $g$, which amounts to two-thirds of the average expenses of students in the benchmark. For students in households higher than the 15 percentile but less than $x$ percentile of the income distribution, where $x$ takes either 40, 50, or 60, the payment is given by $g/2$, which amounts to one-third of the average expenses of students. In other words, the payment tapers off in income. Letting $\bar{I}_{15\%}$ and $\bar{I}_{x\%}$ denote the income level of 15 and $x \in \{40, 50, 60\}$ percentiles of the income distribution, the grant function $\bar{g}(I; x)$ with a threshold $\bar{I}_{x\%}$ can now be formulated as:

$$\bar{g}(I; x) = \begin{cases} g & \text{if } I < \bar{I}_{15\%} \\ g/2 & \text{if } I \in [\bar{I}_{15\%}, \bar{I}_{x\%}) \\ 0 & \text{otherwise} \end{cases} \tag{25}$$

Table 15 summarizes the results, and there are several takeaways there. First, the equilibrium college enrollment increases as the income threshold is higher. Setting the threshold at the 40, 50, and 60 percentiles of the income distribution, the enrollment rate increases by 4.7 p.p., 5.6 p.p., and 6.2 p.p., respectively, in stationary equilibrium. This is also the case for educational mobility and labor income tax rates. The expansion requires additional revenue, so the equilibrium tax rate should also increase by 0.3 p.p. for the case of the 60% threshold.

However, the expansion would not significantly increase the TFR, as Fig 11 indicates. Fertility rates for each skill help us understand this situation. First, college graduates continue to increase fertility. This result is straightforward to understand given that the direct and GE effects contribute to the fertility increase of college graduates when the grants are introduced, as discussed in Sections 7.1 and J.. On the contrary, fertility among high school graduates decreases with expansion, making the TFR remain almost constant even though fertility among college graduates increases. Fig 11 depicts the changes in the TFR and fertility rates for high school and college graduates, where the equilibrium TFR with the existing program (with the 15 percentile threshold) is normalized to one.

| | Threshold | | | |
|---|---|---|---|---|
| | 15% | 40% | 50% | 60% |
| CL share (Δ p.p.) | +4.0 | +4.7 | +5.6 | +6.2 |
| HS→CL (Δ p.p.) | +2.5 | +2.6 | +3.1 | +3.3 |
| Tax (Δ p.p.) | +0.02 | +0.17 | +0.23 | +0.30 |
| | | | | |
| Output (Δ%) | +0.7 | +1.3 | +0.2 | −0.5 |
| Welfare (%) | +5.1 | +6.5 | +6.3 | +5.5 |

Table 15: Main results of higher income thresholds. *Note*: Rows "CL share" and "HS→CL" represent the changes in college enrollment rate and educational mobility in the sense of probability that children of high school graduates attend college. Output changes are expressed as percentage changes. Changes in college enrollment rate, educational mobility, and tax rate are represented as changes in percentage points. Welfare gains are represented in terms of consumption equivalence.

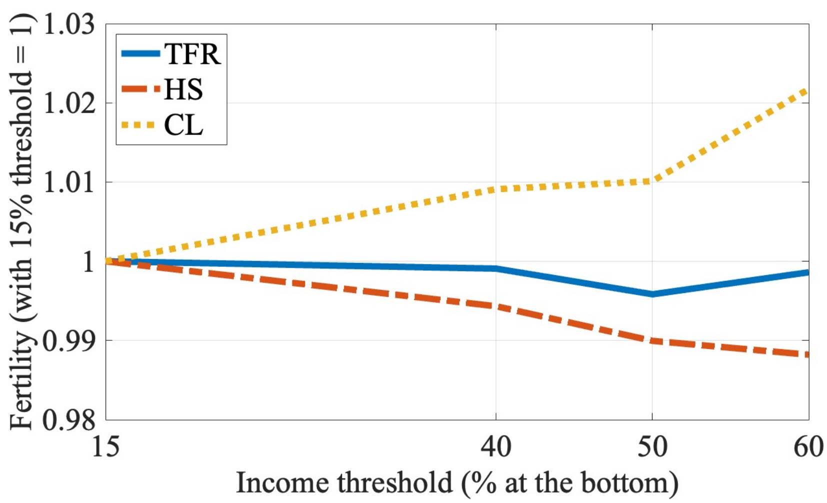


Fig 11: Changes in fertility rates with expansion. *Note*: The equilibrium TFR with the existing program (with the 15% threshold) is normalized to one. "HS" and "CL" represent fertility rates for high school and college graduates.

Why do high school graduates decrease fertility as the income threshold is higher? To understand this, I conduct the decomposition when the grant function is given by $\bar{g}(I;60)$, and the results are summarized in Table 16. Two effects are critical to explain the fertility decreases of high school graduates: the direct and GE effects.

In this case with the grant function $\bar{g}(I;60)$, the wage rate for high school graduates, $w_{HS}$, is 0.6% higher in the long run than in the benchmark, especially because of the significant supply of college graduates. This higher wage rate makes the opportunity costs of having children for high school graduates higher, which puts downward pressure on their fertility rates.

Next, the direct effects imply a lower fertility rate for high school graduates. This result is somewhat confusing given that we discuss the insurance effects of the grants on fertility, which is critical to account for the fertility increases of college graduates. Why do the direct effects lead to lower fertility for high school graduates? In the benchmark without grants, children of high school graduates are unlikely to attend college. If the grants are introduced and their target expands, educational mobility increases in the sense that children of high school graduates are more likely to attend college than in the benchmark, as represented in Table 15. Given that the grants are not generous enough to cover 100% of the costs to send their children to college, this higher probability of children going to college implies that those parents are more likely to have to make additional transfers upon children's college enrollment. The expansion thus increases the expected costs of children for high school graduates, which lowers their fertility.

| | Direct | GE | Tax | Dist. | All |
|---|---|---|---|---|---|
| HS ($\Delta$%) | $-0.7$ | $-1.4$ | 0.0 | 0.0 | $-0.8$ |
| CL ($\Delta$%) | $+9.1$ | $+4.3$ | $+2.4$ | $+0.6$ | $+13.2$ |

Table 16: Decomposing the effects on fertility with $g(h, I, 60)$. *Note*: Rows "HS" and "CL" represent the percentage changes in fertility rates for high school and college graduates.

Last, the output increases in expansion up to the case of the 40 percentile threshold, but they start declining after that point, as Table 15 and Fig 12 indicate. If the income threshold becomes sufficiently high, households with children have less incentive to save to support their college enrollment because they will likely be eligible for the grants. This higher probability of eligibility also causes an income effect on labor supply. Thus, the aggregate capital and labor supply decrease when the income threshold becomes sufficiently high, reducing the output. This concavity of output gains with expansion also applies to the welfare gains. The marginal utility gains of the grants decrease in the values of the income threshold while the tax rate increases with expansion. Then, the net gains start declining at some point.

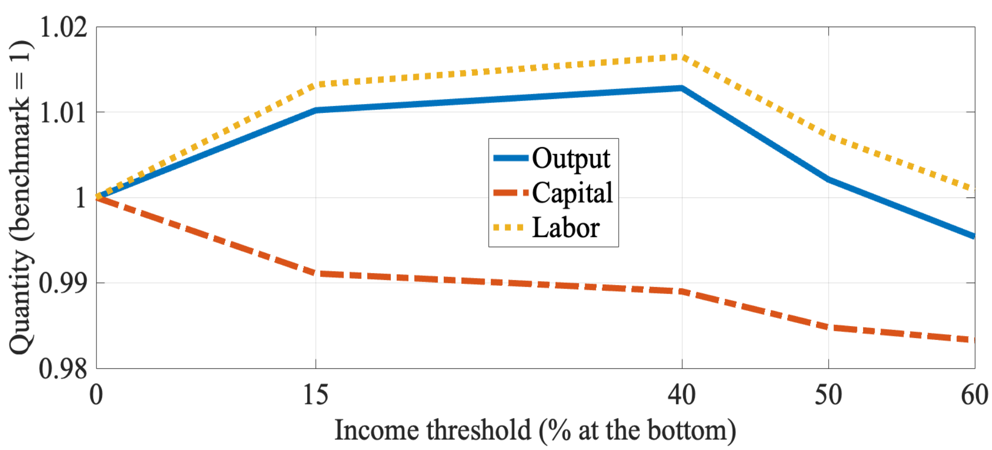


Fig 12: Changes in aggregate output, capital, and labor supply in efficiency unit with expansion. *Note*:

# L. Comparison With Pro-natal Transfers: Macroeconomic Effects

This section examines the differences in the macroeconomic implications between the education subsidy and child benefit (i.e., typical pro-natal cash transfers).[50] To this end,

[50] In Section M., I also consider reallocating resources into different targets to implement the education subsidy by considering unconditional transfers to college students and introducing ability tests.

I simulate the expansion of the unconditional cash transfers to households with children $B$ from its benchmark value while setting $\bar{g}(I) = 0$ for any $I$ as in the benchmark. The new level of the per-child payment $B$ is set so that the short-run expenditure upon the introduction is the same as that of the existing income-tested grants, which leads to a 6.5% increase in the per-child payment.

I solve the stationary equilibrium with the expansion of the per-child payment $B$, and the main results are summarized in Table 17. Columns "Education" and "Typical" represent the results of the introduction of income-tested grants and the expansion of child benefit ("typical" pro-natal transfers).

The child benefit expansion leads to a 2.5% increase of the TFR in the long-run equilibrium, comparable to introducing the income-tested grants but slightly lower than that. The previous decomposition analysis suggests that targeting students in unlucky households in which negative income shocks are realized would be effective, at least marginally, in increasing the TFR. A notable observation is that the characteristics of households who respond to the policy differ between the two cases; the education subsidy induces skilled households to have more children, while the child benefit has a similar degree of impact on skilled and unskilled households regarding fertility. This is because the former is more beneficial for households whose potential children are likely to attend college, and college graduates tend to be those households due to the intergenerational persistence of education.

The college enrollment rate does not change in response to the child benefit expansion. The expansion also does not affect the average human capital, defined as the workers' average of labor efficiency $\eta_{j,z,e,h}$,[51] while the education subsidy increases the average human capital by 1.0% primarily because of the higher college enrollment rate. The difference in the average human capital also leads to the difference in the per-capita output. While the education subsidy increases the per-capita output by 0.7% in the long run, the child benefit expansion would rather lead to a 0.3% lower per-capita output. As in the case of introducing the education subsidy, the child benefit expansion also leads to lower (physical) capital accumulation due to crowding out and demographic change. The education subsidy has sufficiently large positive effects on aggregate labor supply in efficiency units, which surpasses the negative impacts on physical capital accumulation and increases output. The child benefit expansion has more modest impacts on aggregate labor efficiency than the education subsidy, leading to the gap in output gains between the two policies.

The child benefit expansion does not affect the standard deviation of wages, whereas the education subsidy leads to a lower standard deviation by enabling some students in poor households to attend college and acquire skills. Lastly, the expansion leads to a 0.3% of the welfare gain, which is substantially lower than a 4.8% gain with the education subsidy. As discussed in Section J., the welfare gains from the education subsidy primarily stem from higher expected lifetime income across states and an increased probability of attaining a college degree and earning higher wages, driven by higher college enrollment rates and greater educational mobility. In contrast, the expansion of child benefits does

[51] Formally, I define the average human capital as $\sum_j \int_{z,h} [(1-s) \cdot \eta_{j,z,e=0,h} + s \cdot \eta_{j,z,e=1,h}] dF(z,h)\mu_j$, where $s$ represents the college enrollment rate, $F(z,h)$ represents the stationary distribution over $(z,h)$, and $\mu_j$ is the stationary distribution of age.

not yield similar effects. Thus, the education subsidy leads to greater welfare gains than the child benefit expansion under the veil of ignorance.

A caveat is that this model does not capture the endogenous human capital accumulation before high school graduation and the dynamic complementarity of human capital. Households may increase their investments in their children upon the child benefit expansion, which contributes to greater human capital; however, that channel is not considered in this current analysis. Those ingredients can be critical in comparing the effects of these different programs, especially on college enrollment rates, aggregate human capital, and output. The exercise in this framework is a first step, and incorporating those ingredients—fertility, college enrollment, dynamic complementarity of human capital—in one framework is left for future research.

| | Education | Typical |
|---|---|---|
| TFR (Δ%) | +3.1 | +2.5 |
| HS | −0.3 | +2.5 |
| CL | +10.5 | +2.4 |
| CL share (Δ p.p.) | +4.0 | 0.0 |
| HS→CL | +2.5 | 0.0 |
| Avg. HC (Δ%) | +1.0 | 0.0 |
| Output (Δ%) | +0.7 | −0.3 |
| STD (wage) (Δ%) | −1.3 | 0.0 |
| Welfare (%) | +5.1 | +0.3 |

Table 17: Main results with several schemes with different targets. *Note*: Columns "Education" and "Typical" represent the results with the introduction of the education subsidy and expansion of child benefit ("typical" pro-natal transfers), respectively. Values in each cell indicate changes from the benchmark value. Rows "HS" and "CL" indicate the percentage changes in the fertility of high school and college graduates. "HS→CL" represent the changes in college enrollment rate and educational mobility in the sense of probability that children of high school graduates attend college. "Ave. HC" stands for the average human capital.

# M. Reallocating Resources to Different Programs

This section examines the performance of the potential alternative programs to the existing income-tested grants. To this end, I consider another two scenarios, in addition to the introduction of income-tested grants to the benchmark model. The first scenario is to introduce grants for college students with income and ability tests so that "high-ability" students in poor households are eligible. More specifically, I keep the income threshold adopted in the existing scheme but arbitrarily set the lower bound for students' innate ability for the eligibility, $\underline{h}$, to its median. The payment function $g(h, I)$ for this scheme is defined as follows:

$$g(h, I) = \begin{cases} g_1 & \text{if } I < \bar{I} \;\&\; h > \underline{h} \\ 0 & \text{otherwise} \end{cases}$$

Here, the payment $g_1$ is set so that the short-run expenditure upon the introduction is the same as that of the existing income-tested grants. As a result, $g_1$ covers approximately 100% of the average student's expenditure in the benchmark, which is greater than the payment in the existing scheme with only income tests because the number of eligible students is fewer.

The second scenario introduces unconditional grants for college students regardless of ability and income. The payment function is given as $g(h, I) = g_2$ for any students with $(h, I)$, where $g_2$ is set so that the short-run government expenditure is the same as the existing program. As a result, $g_2$ covers approximately 10% of the average students' expenditure in the benchmark, which is less significant than in the existing income-tested grants because this alternative program covers a broader range of students.

I solve the stationary equilibrium with each scenario, and the results regarding fertility and enrollment rates are summarized in Table 18. A highlight is that the existing scheme with income tests would lead to the highest equilibrium TFR among other scenarios. Notably, the unconditional grants lead to a 1.1 p.p. higher college enrollment rate than the income-tested ones and a $-0.3\%$ lower TFR because of the composition effect.

| | Income | +Ability | Uncond. |
|---|---|---|---|
| CL share ($\Delta$ p.p.) | $+4.0$ | $+2.6$ | $+5.1$ |
| TFR ($\Delta$%) | $+3.1$ | $+2.7$ | $-0.3$ |
| HS | $-0.3$ | $-3.5$ | $+4.0$ |
| CL | $+10.5$ | $+8.4$ | $+3.9$ |
| Output ($\Delta$%) | $+0.7$ | $-0.1$ | $-0.7$ |
| STD (wage) ($\Delta$%) | $-1.3$ | $-0.9$ | $-0.1$ |
| Welfare (%) | $+5.1$ | $+2.6$ | $+1.6$ |

Table 18: Main results with several schemes with different targets. *Note*: values in each cell indicate changes from the benchmark value. Rows "HS" and "CL" indicate the percentage changes in the fertility of high school and college graduates.

# N. Other Policies Insuring against the Risks

This section studies two other policies that potentially insure against the risks associated with fertility choices under income uncertainty, other than the income-tested subsidy examined in Section 7: (1) wage insurance that provides cash benefits with households with a negative income shock and (2) fertility-dependent wage insurance that provides wage insurance benefits with households depending on the number of children they have.

Wage insurance provides workers facing negative income shocks (i.e., $z < 0$) with cash transfers. In the uniform wage insurance, I introduce the following payment function into the budget constraint for workers:

$$w_I(z) = \begin{cases} w_I & \text{if } z < 0 \\ 0 & \text{otherwise} \end{cases}$$

For the fertility-dependent wage insurance, I introduce the following payment function into the budget constraint for workers:

$$\tilde{w}_I(z, n, j) = \begin{cases} n \cdot \tilde{w}_I & \text{if } z < 0 \ \&\ j < J_{IVT} \\ 0 & \text{otherwise} \end{cases}$$

Under the fertility-dependent wage insurance, workers facing a negative income shock receive wage insurance benefits proportional to the number of children younger than 18 that they have.

I implement the introduction of these policies in an expenditure neutral way. More specifically, I feed the income-tested education subsidy into the model as I do in Section 7, and then set policy parameters for each cash transfer so that the short-run government expenditures of each policy is the same as those of the income-tested education subsidy. Also, I simulate each policy in a partial equilibrium setup to focus on the direct effects on fertility, controlling for indirect effects such as a GE effect and distributional effect.

The results are summarised in Table 19. Introducing fertility-dependent wage insurance increases fertility by 0.9%, whereas introducing uniform wage insurance has no effect on fertility. The insurance effects of the income-tested college subsidy on fertility, which result in a 1.3% increase in fertility, are more significant than those of these policies. This is because the income-tested college subsidy provides partial insurance not only against income risk but also against another source of uncertainty: uncertainty regarding children's characteristics (i.e., psychic costs and innate abilities) and the associated uncertainty regarding the willingness to pay for children's education.

| | Fertility ($\Delta$%) |
|---|---|
| Income-test college subsidy | +2.1% |
| Wage insurance (uniform) | +0.0% |
| Wage insurance (fertility) | +0.9% |

Table 19: Changes in fertility with policies that insure against the risks.

# O. Transitional Dynamics: Additional Figures

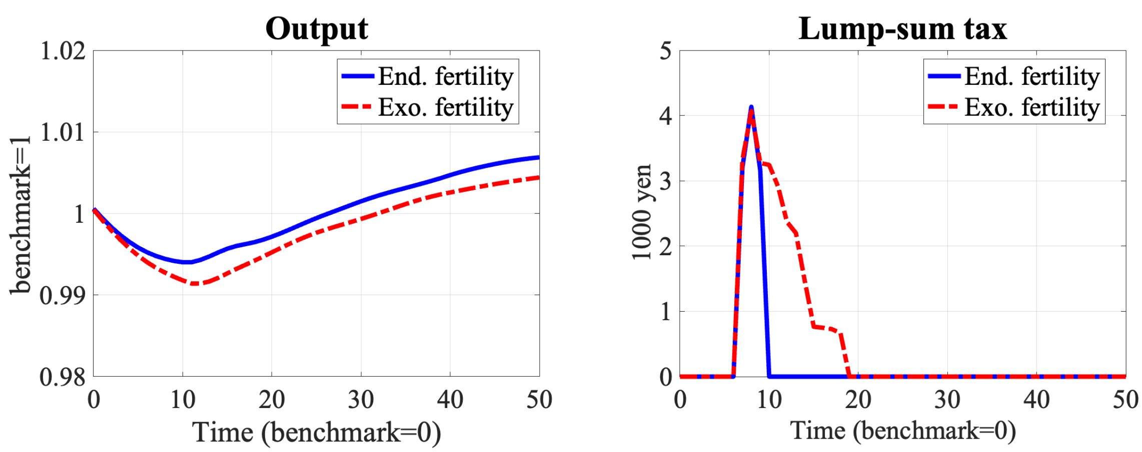


Fig 13: Transitional dynamics of output and lump-sum tax. *Note*: "End. fertility" and "Exo. fertility" represent transition paths under endogenous fertility and exogenous fertility.

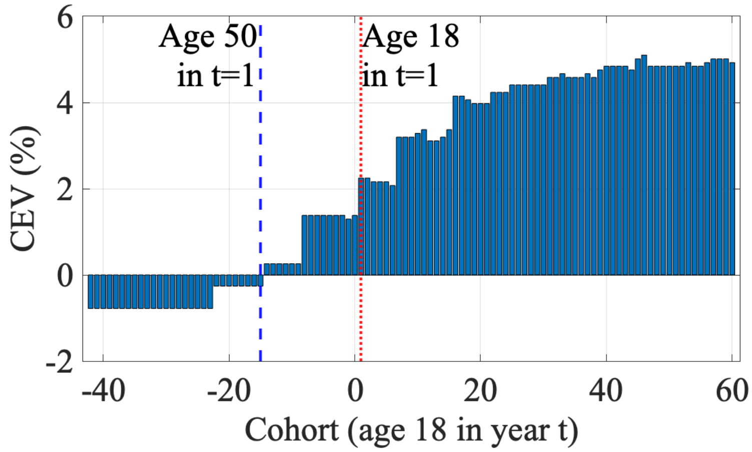


Fig 14: Welfare effects for each cohort along the transition. *Note*: The horizontal axis represents the cohort, where larger values mean younger cohorts.